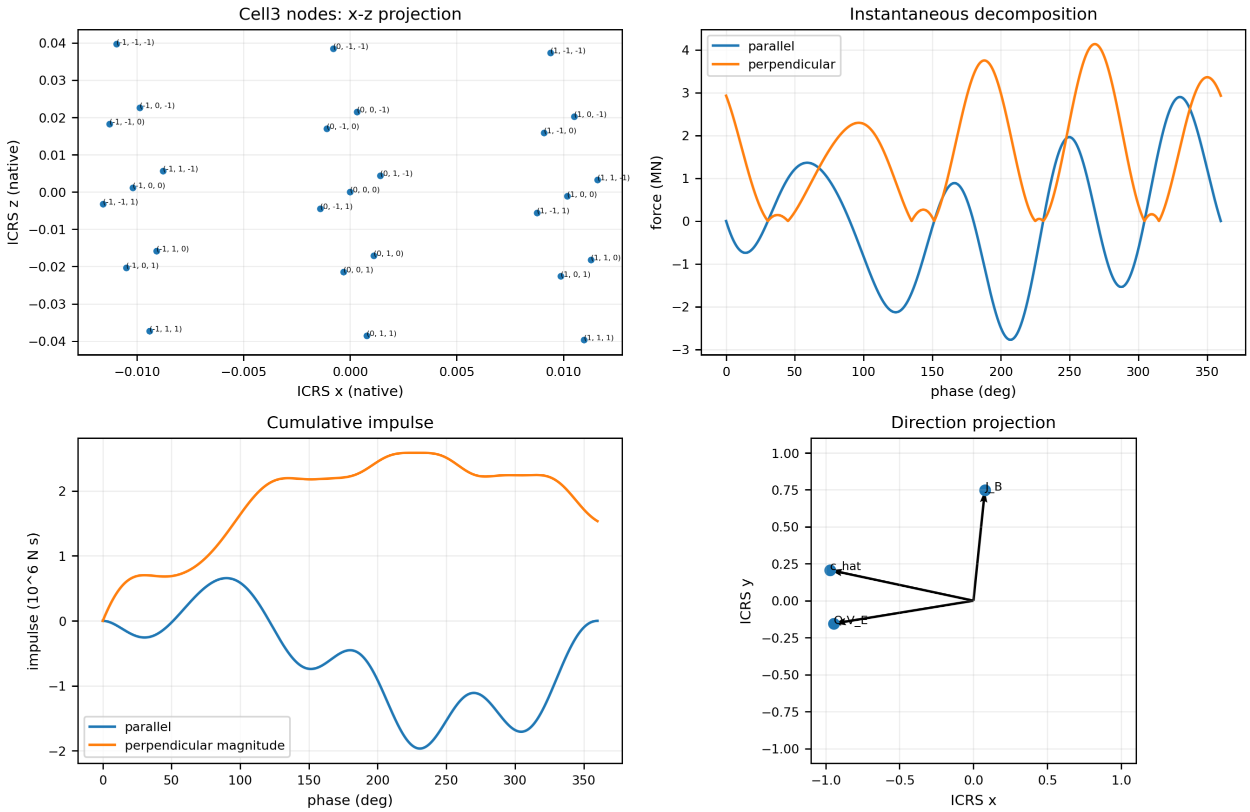
K-FIELD DUAL-AXIS KINETIC MASS CMB TRAVERSAL NET-FORCE VECTOR

Instantaneous axial and transverse force geometry, closed-cycle orthogonality, and Cell3 traversal in the Earth-CMB frame

K-Field dual-axis CMB traversal net-force geometry



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Frontispiece: authoritative base cell, face planes, and all nodes

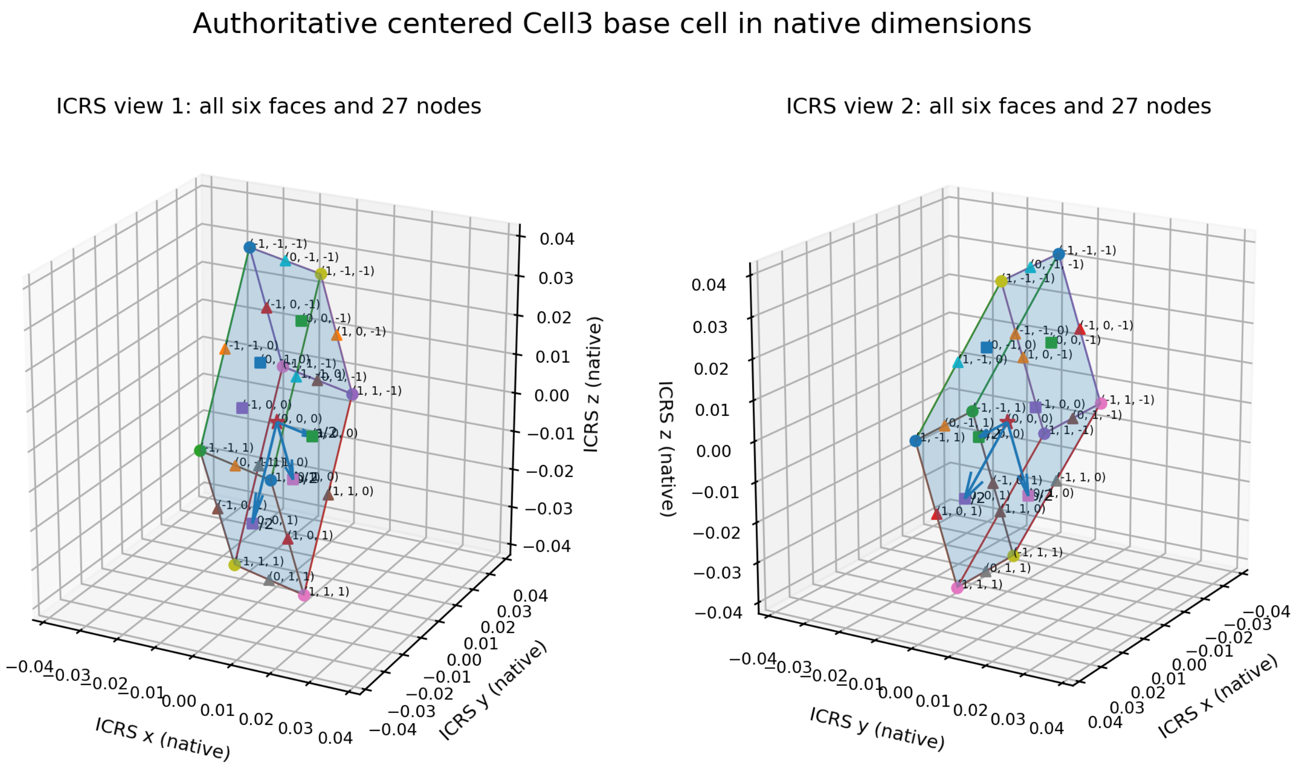

Figure 1. Authoritative centered Cell3 base cell in fixed ICRS coordinates. Both views use equal native-length scaling. The six quadrilateral boundary faces, 12 edges, all 27 centered half-step nodes, integer node addresses, and basis half-vectors are rendered directly from the source basis matrix.

Abstract

This reference isolates the directional question that governs interpretation of the dual-axis kinetic-mass traversal: whether the system net-force vector remains perpendicular to the Earth-CMB traversal. The answer has two distinct levels. The instantaneous net force is not generally perpendicular. It is radial along the moving endpoint direction and therefore carries a time-varying component parallel to the CMB axis. At the four equatorial phases of the local spherical path the force is exactly transverse; at the four CMB-axis centerline phases it is exactly axial. Four additional phases make the complete force vanish because the Q-metric translation-rotation cross scalar is zero. The complete force integral over one outer cycle is different from any single instantaneous vector. Harmonic orthogonality converts the full vector sum into J_B = pi kappa_F m c_hat cross (Q V_E), which is exactly perpendicular to c_hat. The cycle-average force is therefore also exactly transverse. The analysis is evaluated for two 1 kg endpoint masses at R = 0.5 m, outer rate omega_B = 1 rad/s, inner rate omega_A = 2 rad/s, and kappa_F = 1. The continuous maximum force magnitude is 4.145151 x 10^6 N in the native Q-metric normalization. The maximum instantaneous axial magnitude is 2.897630 x 10^6 N. The exact closed-cycle impulse is (118842.532043, 1146514.977540, 1012275.514802) N s with magnitude 1.534054060 x 10^6 N s. Its dot product with the CMB unit axis is 1.294 x 10^-11 N s, a relative orthogonality residual of 8.433 x 10^-18 from floating-point rendering. The RA = 0 degree seasonal datum changes only the absolute position origin; it does not alter these force-direction relations.

```
Instantaneous: F_net dot c_hat is generally nonzero.
Closed cycle: c_hat dot J_B = 0 exactly.
Cycle average: c_hat dot <F_net> = 0 exactly.
```


Keywords: K-Field, Cell3, Earth-CMB velocity, dual-axis rotation, instantaneous force, axial force, transverse force, closed-cycle impulse, vector orthogonality.

Contents

1. Purpose, question, and central answer

The device is placed at Earth center. Axis B is locked to the Earth-CMB traversal direction, while axis A is perpendicular to axis B and rotates with the assembly. The endpoint pair remains antipodal on a rigid massless bar. The selected seasonal position datum places Earth center at RA = 0 degrees and Dec = 0 degrees in ICRS, at one astronomical unit on the positive ICRS x axis. This datum fixes a coordinate origin for the traversal but does not rotate the apparatus frame or the CMB direction.

The phrase net force can refer either to the force vector at one instant or to the vector obtained by integrating and averaging the full cycle. These are not interchangeable. The instantaneous vector continuously changes direction. The cycle integral adds all instantaneous vectors with their signs and durations. The central conclusion is therefore stated in two parts.

```
F_net(phi) is not generally perpendicular to c_hat.
J_B = integral F_net dt is exactly perpendicular to c_hat.
<F_net> = J_B/T_B is exactly perpendicular to c_hat.
```

The transverse final result does not mean that every contributing force vector was transverse. It means that the complete set of axial contributions cancels over the closed 2:1 trajectory while a transverse vector residue remains.

2. Apparatus, coordinates, and declared datum

All vectors are expressed in the fixed ICRS Cartesian frame. The Earth-CMB unit direction is also the outer rotation axis c_hat. The vectors e0, A0, and c_hat form an orthonormal right-handed frame. The apparatus values are listed below.

Quantity	Definition	Value
Endpoint masses	Two equal point masses	1 kg each
Bar	Rigid and massless	1 m total length
Mass radius	Half the bar length	0.5 m
Axis B	Earth-CMB unit direction	(-0.970770855836, 0.207348009111, -0.120874929482)
Axis A at phase zero	Perpendicular reference	(0.131366829113, 0.880530254232, 0.455422032396)
Bar reference e0	Endpoint direction at phi = 0	(-0.200864884110, -0.426231479950, 0.882031758970)
Rate lock	omega_A / omega_B	2
Numerical normalization	omega_B; kappa_F	1 rad/s; 1

The Earth-CMB velocity vector is V_E = (-359003.199066, 76679.886034, -44701.060101) m/s, with speed 369812.502000 m/s. Its direction is c_hat. At the RA = 0 degree seasonal datum, the Earth-center position is r_E,0 = (149597870700, 0, 0) m. The center then advances locally as r_C(phi) = r_E,0 + (V_CMB/omega_B) phi c_hat. Because the force law depends on velocity and local rotor phase, adding this absolute position translation does not rotate or otherwise alter the force decomposition.

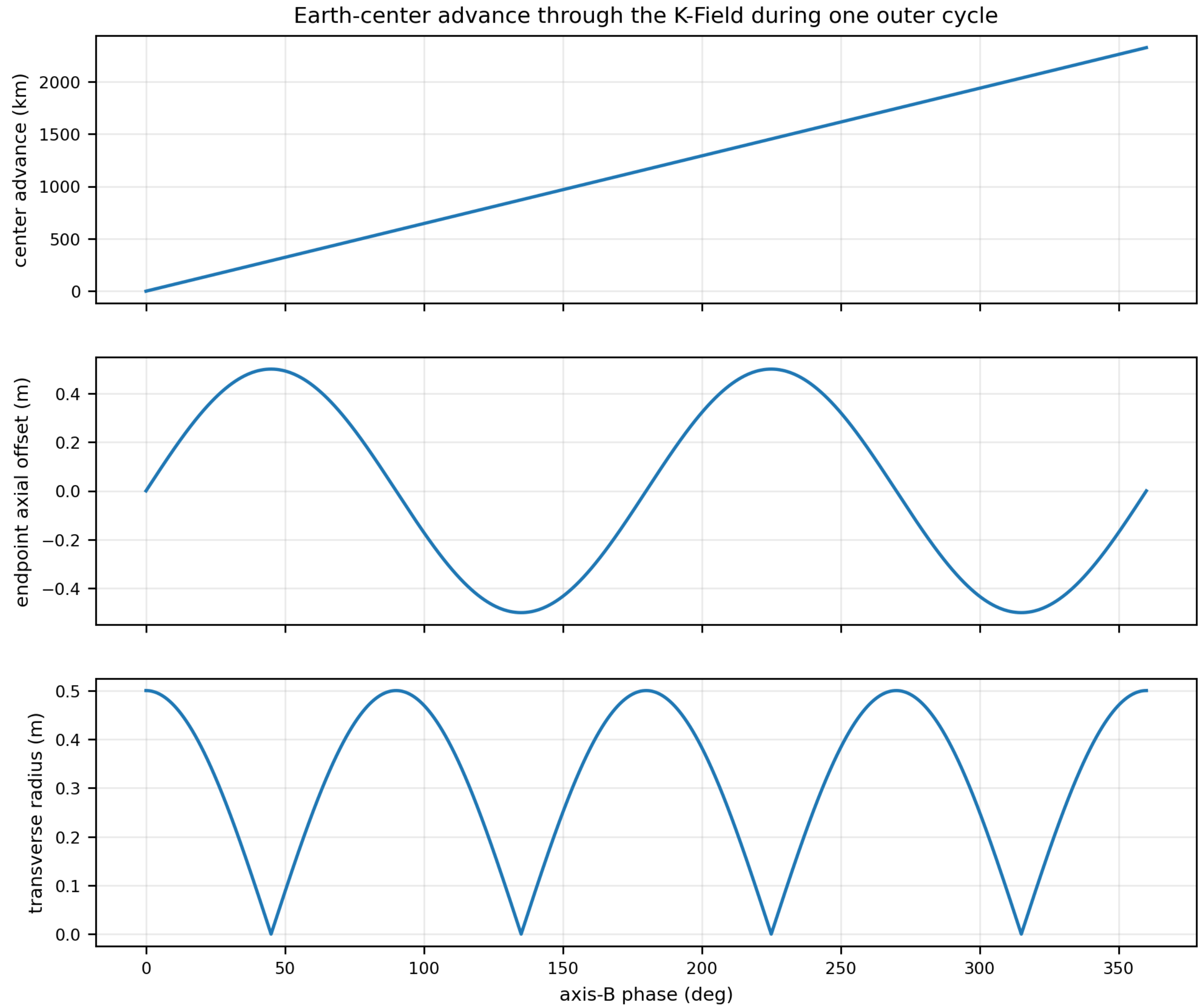

Figure 2. Dimensionally separated components of the absolute traversal during one unit-rate outer cycle. The center advances 2323.600479 km along the CMB direction while the endpoint remains within a 0.5 m local axial and transverse envelope. The separate panels prevent the kilometer-scale translation from visually suppressing the meter-scale rotor geometry.

3. Cell3 structure and oblique CMB traversal

The K-Field base structure is represented by the centered primitive Cell3 generated from the three ICRS basis vectors a, b, and c. The 27 half-step nodes consist of eight corners, twelve edge centers, six face centers, and one body center. The six boundary faces form three opposite plane pairs. The Earth-CMB direction is not parallel to any Cell3 edge or face normal; it crosses all three plane families obliquely.

Basis	ICRS x	ICRS y	ICRS z	Length
a	0.020381438517	-0.003416414496	-0.002332965917	0.020797058782
b	0.002198745968	0.012902170259	-0.034066501536	0.036494205131
c	-0.000630379485	-0.024257722017	-0.042924354571	0.049308565900

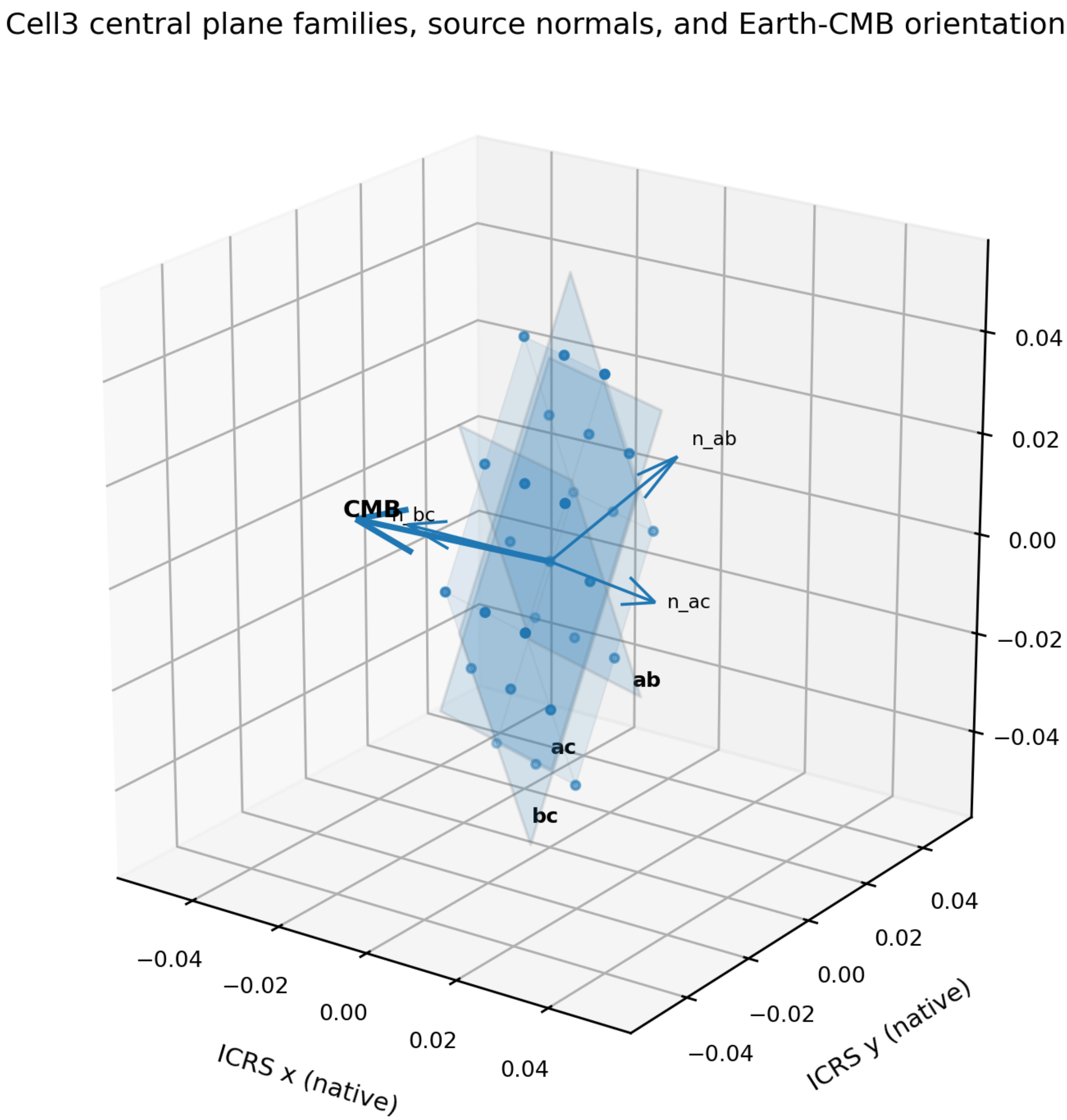

Figure 3. Dimensionally correct native Cell3 central plane families with source face normals and the Earth-CMB orientation vector. The orientation arrow indicates direction; the cell, nodes, and planes retain native-length scaling.

In Cell3 coefficient coordinates, the CMB direction is B_K^-1 c_hat = (-48.1810381420, 5.4293049531, 1.1257530662). Thus the path crosses the constant-a plane stack most rapidly. The native path lengths between consecutive coefficient-plane crossings are 0.020755053 for the bc family, 0.184185639 for the ac family, and 0.888294272 for the ab family. These are native lengths along the CMB-directed line. A physical meter conversion requires an independent scale bridge and is not introduced here.

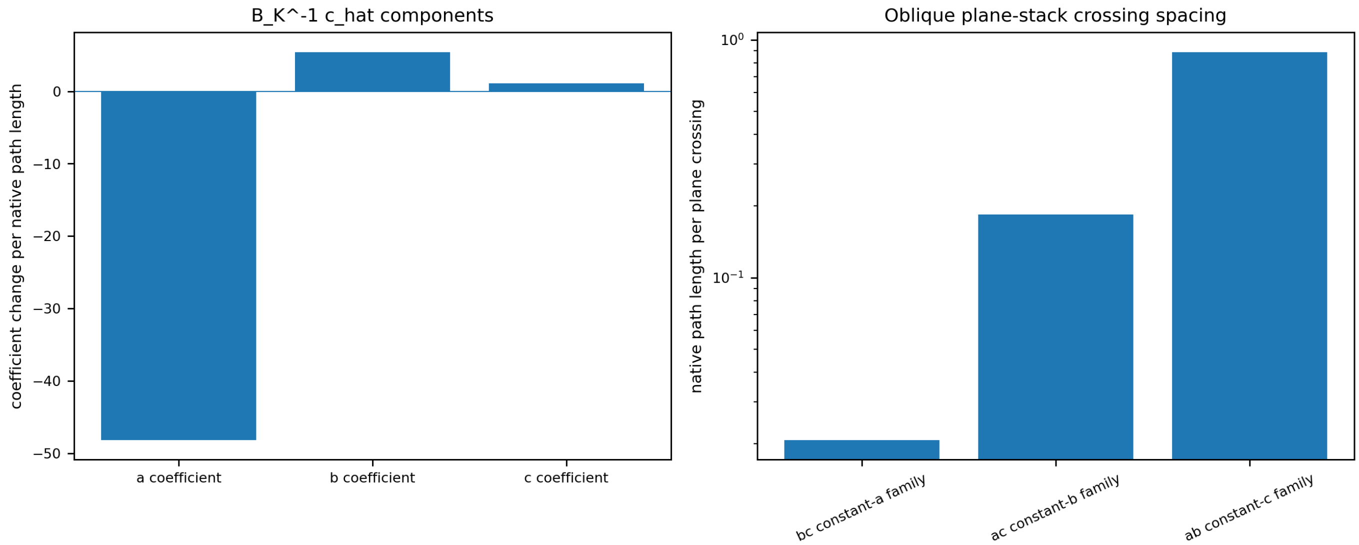

Figure 4. Cell3 coefficient traversal. Left: coefficient-rate components of B_K^-1 c_hat. Right: native path length required to cross one member of each translated plane stack. The logarithmic axis preserves the large spacing ratio without changing the underlying values.

4. Exact 2:1 compound-rotation path

Let phi = omega_B t be the outer mechanical phase. The transverse vector e(phi) rotates once around c_hat while the bar phase advances at 2 phi. The positive endpoint direction is u(phi); the negative endpoint is -u(phi).

```
e(phi) = cos(phi)e0 + sin(phi)A0
u(phi) = cos(2phi)e(phi) + sin(2phi)c_hat
r_plus(phi) = r_C(phi) + R u(phi)
r_minus(phi) = r_C(phi) - R u(phi)
```

Because e(phi) is perpendicular to c_hat, u(phi) has unit length. Each endpoint therefore remains on a radius-0.5 m sphere centered on the translating Earth center. The fixed-frame harmonic expansion contains first-, second-, and third-harmonic terms:

```
u(phi) = 0.5[cos(3phi)+cos(phi)]e0
+ 0.5[sin(3phi)-sin(phi)]A0
+ sin(2phi)c_hat
```

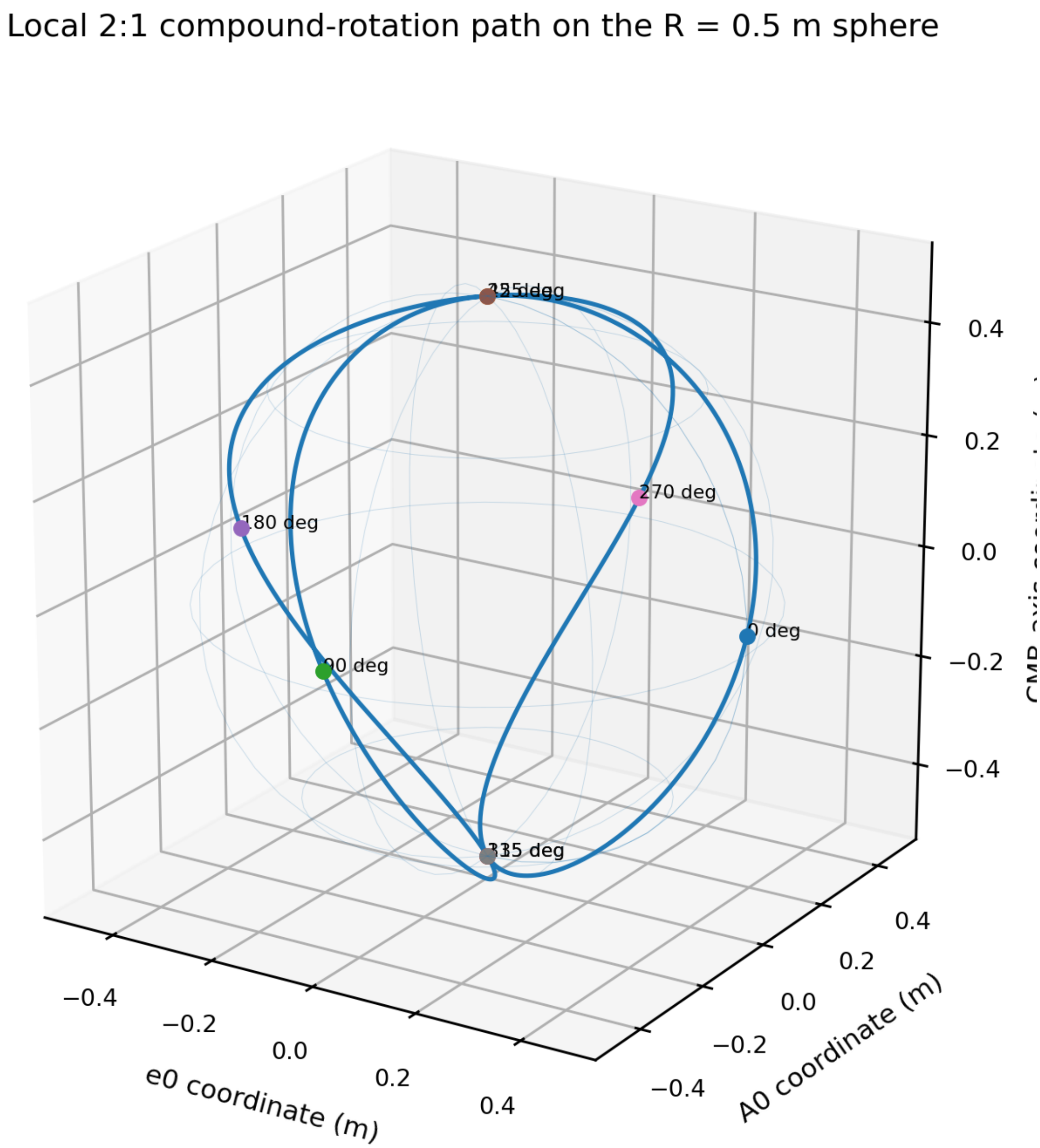

Figure 5. The exact local endpoint path on the 0.5 m sphere in the orthonormal (e0, A0, c_hat) frame. The path has four equatorial states at 0, 90, 180, and 270 degrees and four CMB-axis centerline states at 45, 135, 225, and 315 degrees.

5. Q-metric force law and system net force

The endpoint velocities in the K-Field frame contain the common Earth-CMB translation and equal opposite rotational velocities: v_plus = V_E + R omega_B u_phi and v_minus = V_E - R omega_B u_phi. The Q-metric energy is one-half m kappa_F v^T Q v. Its radial derivative produces the endpoint force channels. When the two endpoint vectors are added, the equal pure-rotation terms cancel and the translation-rotation cross terms reinforce.

```
B(phi) = u_phi^T Q V_E
C(phi) = u_phi^T Q u_phi
F_plus = -kappa_F m[omega_B B + R omega_B^2 C]u
F_minus = kappa_F m[R omega_B^2 C - omega_B B]u
F_net(phi) = -2 kappa_F m omega_B B(phi)u(phi)
```

The force vector is always collinear with u(phi), although its sign reverses when B(phi) changes sign. It is not constrained to the plane perpendicular to c_hat. Its direction follows the radial endpoint direction on the local sphere.

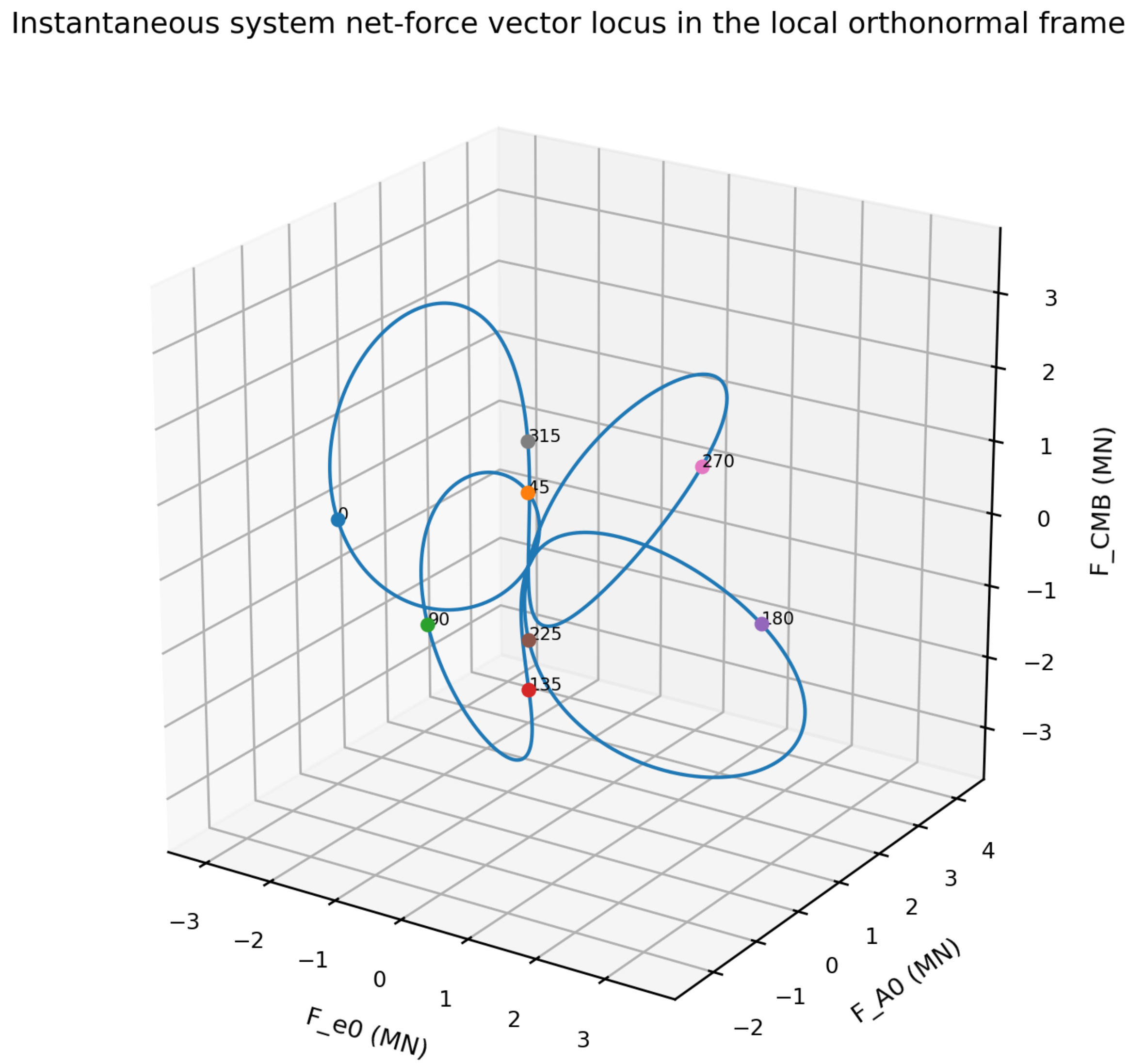

Figure 6. Locus of the instantaneous net-force vector in the local orthonormal component space. Axes are force components in meganewtons under the declared native normalization. The locus is not confined to the plane F_CMB = 0.

6. Instantaneous axial and transverse decomposition

Any instantaneous force vector can be decomposed uniquely into a component parallel to the CMB axis and a component perpendicular to it. Since u dot c_hat = sin(2phi), the axial scalar is obtained directly.

```
F_parallel_scalar(phi) = F_net(phi) dot c_hat
= -2 kappa_F m omega_B B(phi) sin(2phi)
F_parallel(phi) = F_parallel_scalar(phi)c_hat
F_perp(phi) = F_net(phi) - F_parallel(phi)
= -2 kappa_F m omega_B B(phi) cos(2phi)e(phi)
```

Therefore the instantaneous force is perpendicular to the CMB traversal exactly when B(phi) sin(2phi) = 0. This condition contains two geometrically different cases. When sin(2phi) = 0, the nonzero endpoint radius lies in the equatorial plane and the force is genuinely transverse. When B(phi) = 0, the complete force vector vanishes, so its dot product is zero but its direction is undefined.

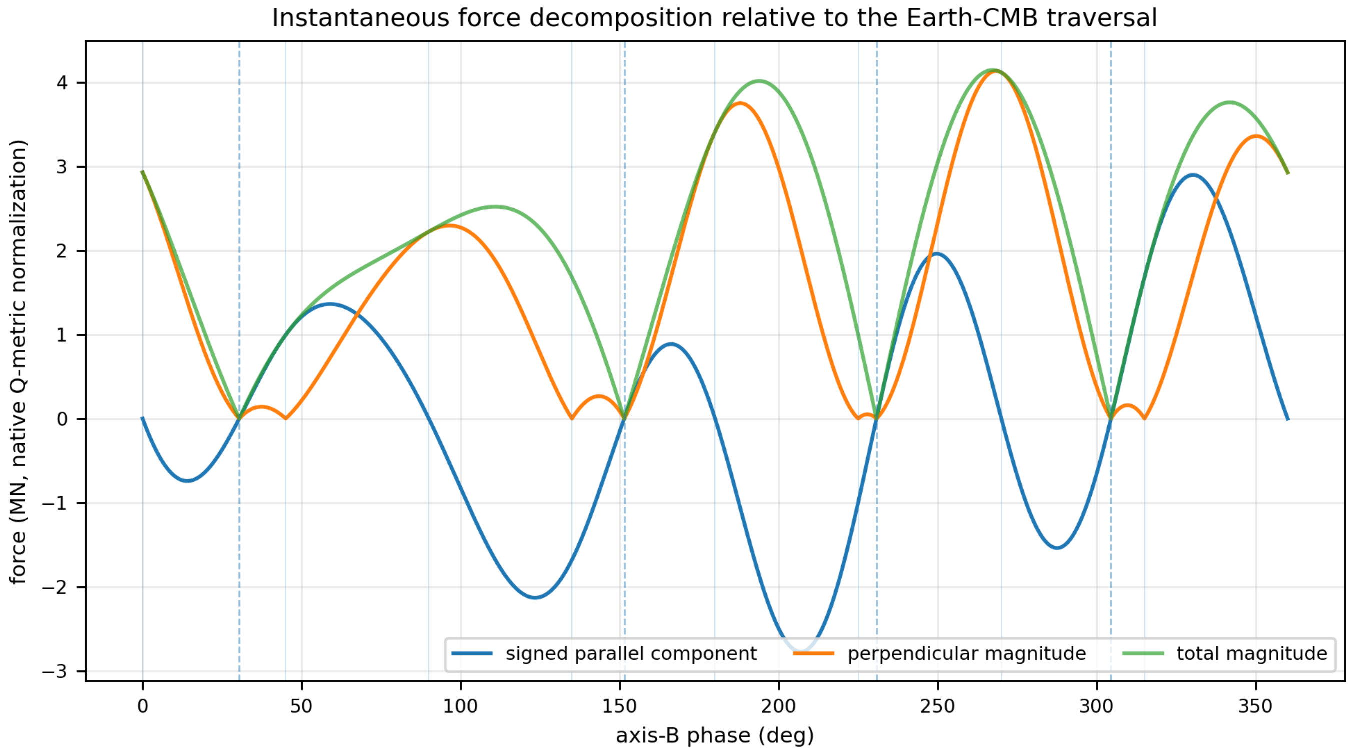

Figure 7. Signed parallel force, perpendicular magnitude, and total magnitude over one outer cycle. Vertical reference lines identify the 45-degree geometry sequence; dashed lines identify the four B(phi) = 0 zero-force phases. The instantaneous vector is mixed axial-transverse during most of the cycle.

The continuous maximum total force is 4.145151e+06 N at phi = 267.246 degrees. The maximum absolute axial component is 2.897630e+06 N at phi = 330.264 degrees. The maximum perpendicular magnitude is 4.133570e+06 N at phi = 268.344 degrees.

7. Phase classes: transverse, axial, mixed, and zero-force

The 2:1 geometry supplies a simple phase classification independent of the Cell3 scale. At multiples of 90 degrees, u lies in the plane perpendicular to c_hat and the force is purely transverse. At odd multiples of 45 degrees, u lies on the CMB axis and the force is purely axial. At all other nonzero-force phases, both components are present.

Phase (deg)	Endpoint geometry	Force geometry	|F| (N)	F parallel (N)	|F perp| (N)
0	equatorial radius: u perpendicular to CMB axis	purely transverse to CMB traversal	2.927732e+06	1.701513e-11	2.927732e+06
45	CMB-axis centerline: u parallel or antiparallel to CMB axis	purely axial along the CMB traversal line	9.956225e+05	9.956225e+05	1.301563e-10
90	equatorial radius: u perpendicular to CMB axis	purely transverse to CMB traversal	2.223721e+06	2.719781e-10	2.223721e+06
135	CMB-axis centerline: u parallel or antiparallel to CMB axis	purely axial along the CMB traversal line	1.680416e+06	-1.680416e+06	3.406516e-10
180	equatorial radius: u perpendicular to CMB axis	purely transverse to CMB traversal	3.411954e+06	1.031255e-09	3.411954e+06
225	CMB-axis centerline: u parallel or antiparallel to CMB axis	purely axial along the CMB traversal line	9.956225e+05	-9.956225e+05	3.334263e-10
270	equatorial radius: u perpendicular to CMB axis	purely transverse to CMB traversal	4.115966e+06	1.564650e-09	4.115966e+06
315	CMB-axis centerline: u parallel or antiparallel to CMB axis	purely axial along the CMB traversal line	1.680416e+06	1.680416e+06	8.435091e-10
360	equatorial radius: u perpendicular to CMB axis	purely transverse to CMB traversal	2.927732e+06	1.291821e-09	2.927732e+06

The four Q-metric zero-force phases are distinct from the geometric transverse phases. They occur at the roots of B(phi):

Root	Phase (deg)	Phase (rad)
1	30.402363199	0.530621338217
2	151.435410918	2.643046524636
3	230.733949264	4.027067110793
4	304.459020416	5.313812343653

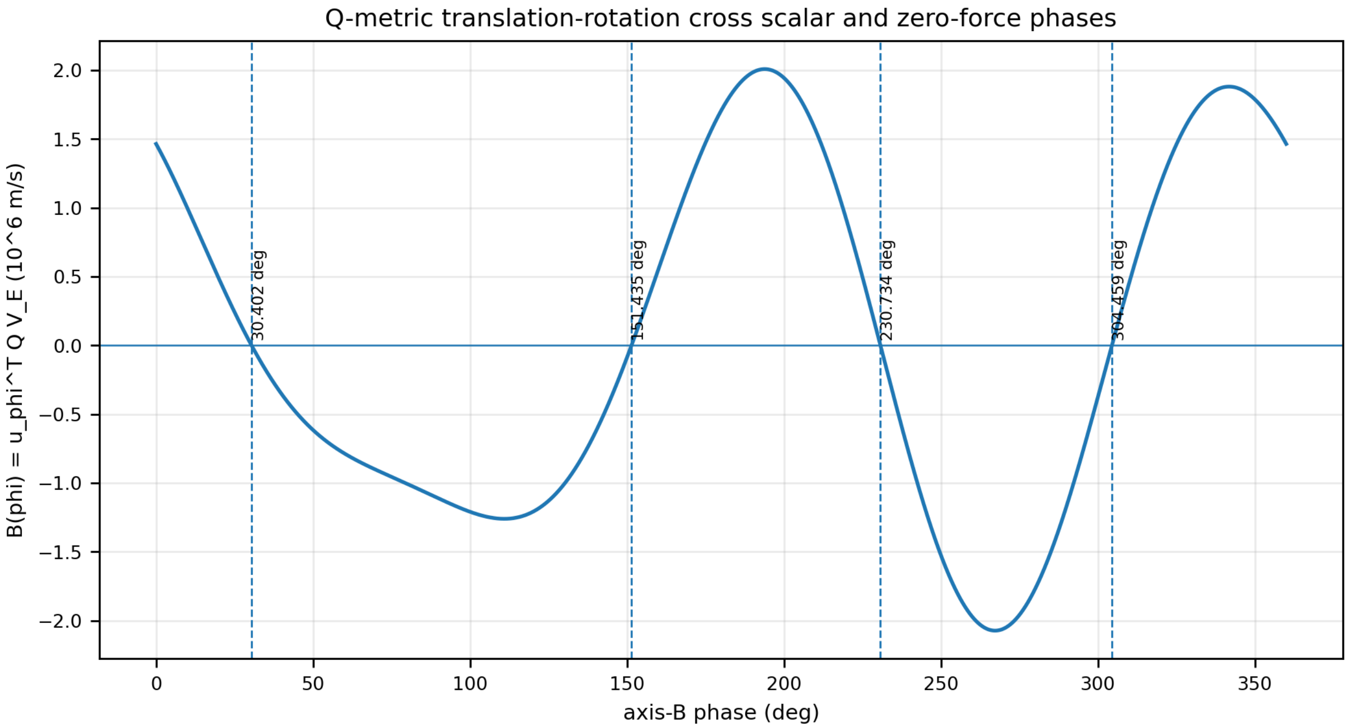

Figure 8. Q-metric cross scalar B(phi) and its four exact numerical roots. At these roots F_net = 0 because the scalar multiplying u(phi) vanishes.

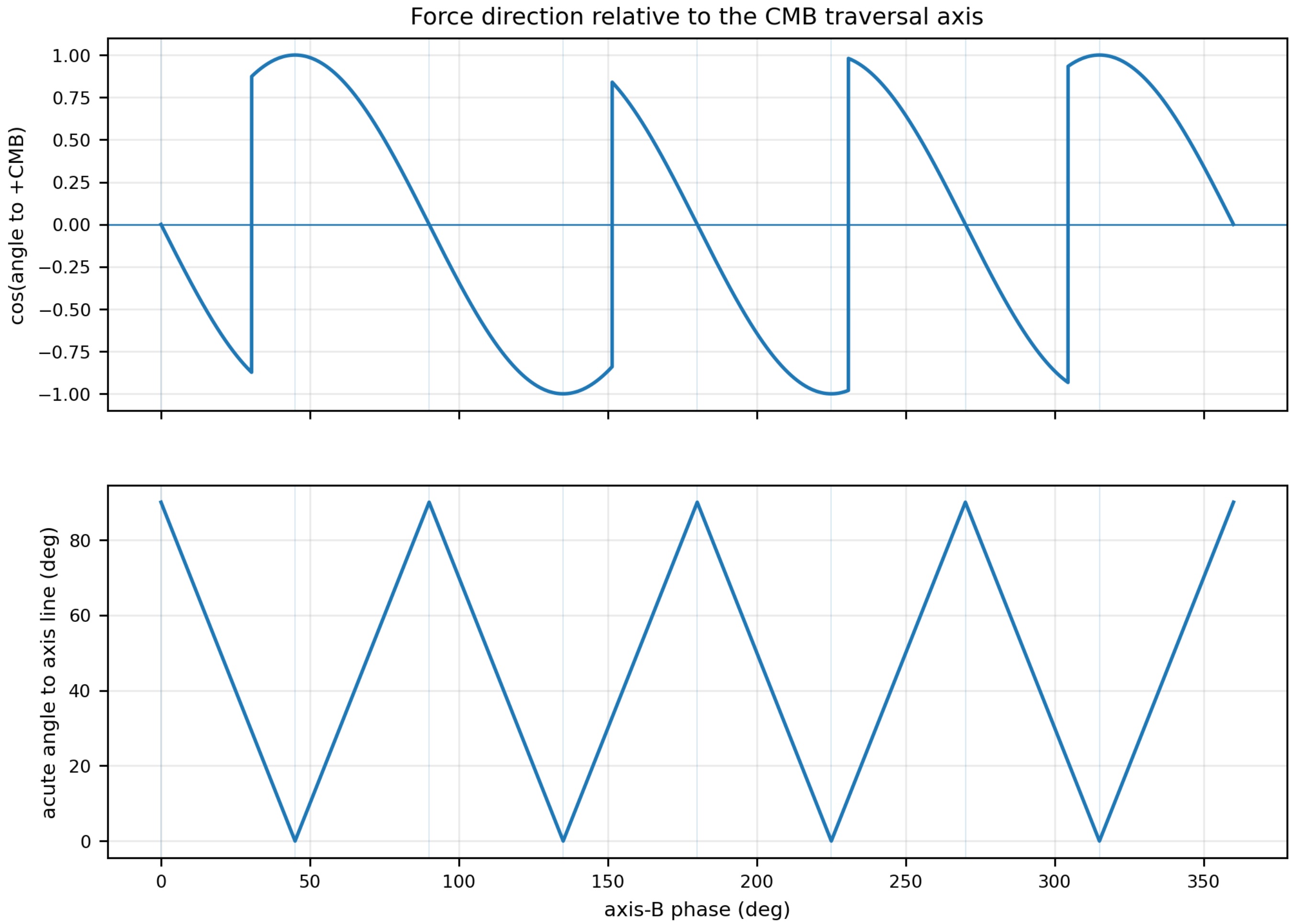

Figure 9. Direction angles. The upper panel gives the signed cosine relative to the positive CMB vector. The lower panel gives the acute angle to the unoriented CMB axis line. A value of 90 degrees is transverse; a value of 0 degrees is axial.

8. Why the closed-cycle result is perpendicular

The closed-cycle impulse is the time integral of the complete instantaneous force waveform. With dphi = omega_B dt, the explicit factor omega_B cancels from the integral:

```
J_B = integral_0^T_B F_net(t) dt
= -2 kappa_F m integral_0^(2pi) [u_phi^T Q V_E]u dphi
```

Define the dyadic integral M = integral u u_phi^T dphi. Substitution of the first-, second-, and third-harmonic expansion and orthogonality of unequal sine and cosine harmonics gives the exact skew-symmetric matrix

```
M = [[0, pi/2, 0],[-pi/2, 0, 0],[0, 0, 0]]
= -(pi/2)[c_hat cross]
J_B = -2 kappa_F m M(Q V_E)
= pi kappa_F m c_hat cross (Q V_E)
```

A cross product with c_hat is perpendicular to c_hat by definition. Consequently c_hat dot J_B = 0 exactly. Dividing by the period does not change direction, so c_hat dot = 0 exactly. This is a closed-cycle cancellation theorem, not an instantaneous constraint.

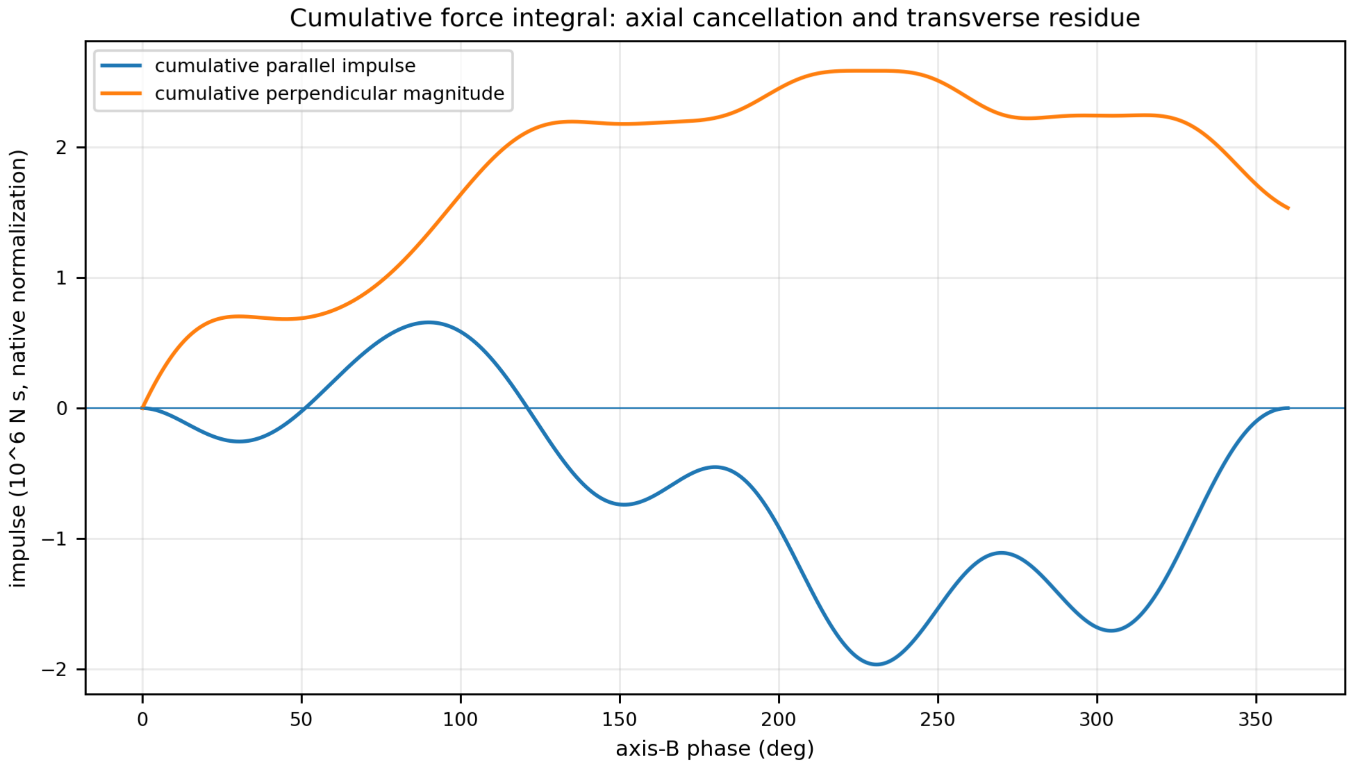

Figure 10. Cumulative impulse decomposition. The intermediate cumulative impulse develops a substantial axial component, reaching an absolute maximum of approximately 1.964633 x 10^6 N s. That axial component returns to zero at cycle closure, while the perpendicular residue terminates at the analytic impulse magnitude.

9. Numerical closed-cycle vector and orthogonality

Quantity	Value	Unit
Analytic impulse x	1.188425320426e+05	N s
Analytic impulse y	1.146514977540e+06	N s
Analytic impulse z	1.012275514802e+06	N s
Impulse magnitude	1.534054060003e+06	N s
Impulse dot CMB axis	1.293619088505e-11	N s
Relative orthogonality residual	8.432682538597e-18	dimensionless
Angle to positive CMB vector	90.000000000000	deg
Cycle-average magnitude	2.441522866197e+05	N at omega_B = 1

The floating-point dot product is 1.294 x 10^-11 N s. Relative to the impulse magnitude, this is 8.433 x 10^-18. The analytic form is exactly orthogonal; the nonzero decimal is the residual of finite numerical rendering. The one-degree trapezoidal integration reproduces the analytic vector to better than 10^-6 N s in this independently regenerated report.

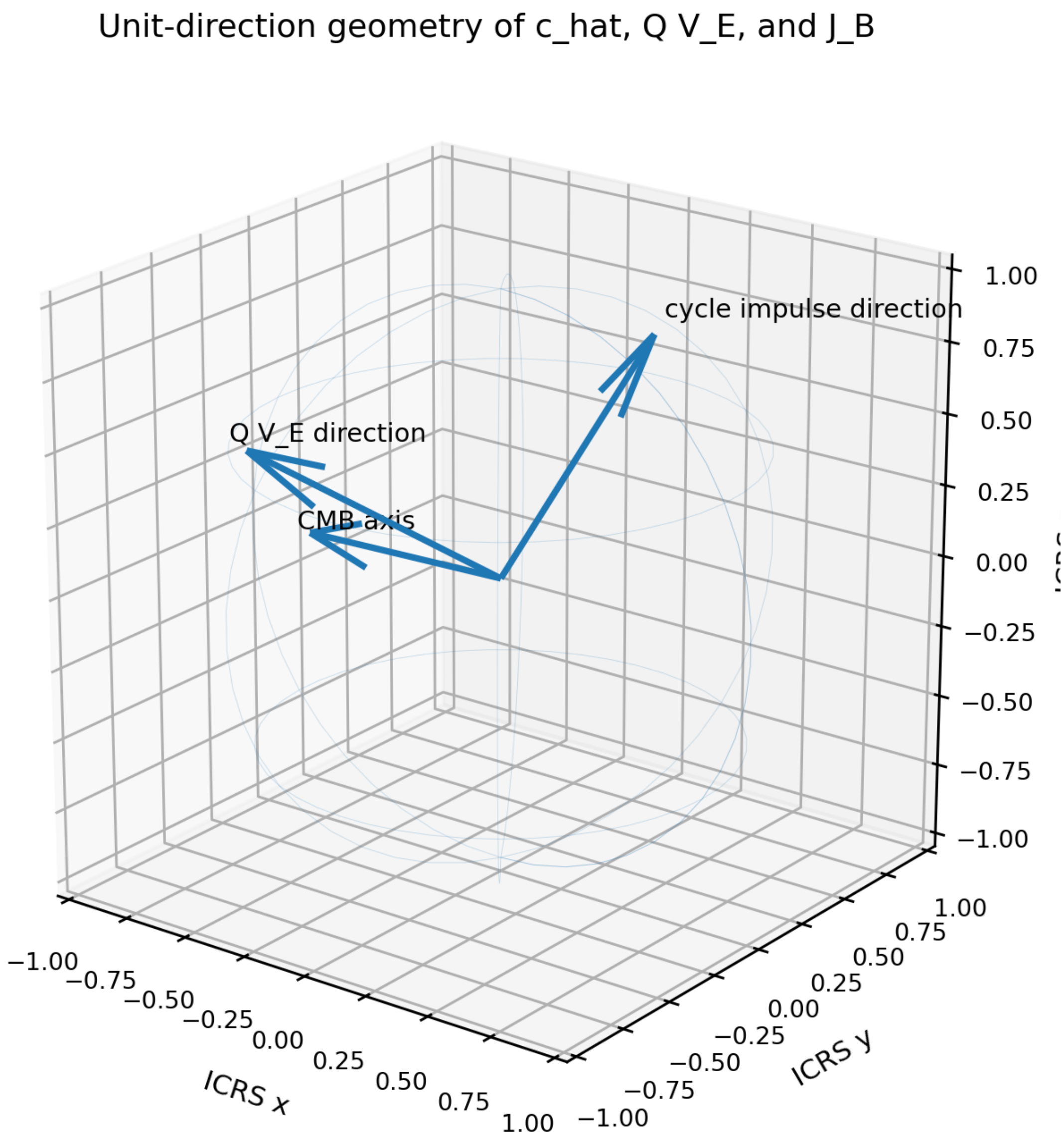

Figure 11. Unit-direction geometry in ICRS. The cycle impulse direction is c_hat cross (Q V_E), so it lies in the plane perpendicular to c_hat. Q V_E itself is not generally parallel to V_E because Q is anisotropic.

10. Effect of the RA = 0 degree seasonal datum

The selected seasonal datum assigns the Earth center a position on the positive ICRS x axis. It is an origin and phase reference, not an additional force direction. Translating the complete endpoint path by r_E,0 adds the same constant position to every point. Translating the center uniformly by V_E t adds the same common centerline motion already represented in the endpoint velocities. Neither operation changes u(phi), u_phi, B(phi), the decomposition relative to c_hat, or the cross-product impulse theorem.

```
r_plus(phi) = r_E,0 + (V_CMB/omega_B)phi c_hat + R u(phi)
F_net(phi) is unchanged by the constant datum r_E,0.
J_B and <F_net> remain perpendicular to c_hat.
```

Placing the device at Earth center also removes surface-latitude displacement and diurnal rotational velocity from this datum. A future extension that explicitly adds Earth orbital velocity or other time-dependent translations must recompute the velocity vector used in Q V. That extension is outside the declared locally frozen RA = 0 degree datum and is not silently included here.

11. Scaling laws and interpretation

The instantaneous net-force magnitude is proportional to kappa_F m omega_B. Its direction at a specified phase is determined by the sign of B(phi) and the geometric radial direction u(phi). The impulse per outer cycle is proportional to kappa_F m but independent of R and omega_B for the declared radial force law. The cycle-average force is proportional to kappa_F m omega_B.

Quantity	Magnitude scaling	Directional statement
Instantaneous F_net	kappa_F m omega_B	Direction varies with phase
Instantaneous F_parallel	kappa_F m omega_B	Generally nonzero
Instantaneous F_perp	kappa_F m omega_B	Generally nonzero
Impulse per outer cycle	kappa_F m	Exactly transverse; independent of R and omega_B
Cycle-average force	kappa_F m omega_B	Exactly transverse
Individual pure-rotation terms	kappa_F m R omega_B^2	Cancel in the two-mass system sum

The essential geometric distinction is between local direction and cycle closure. The local path alternates among transverse, mixed, and axial force states. The complete cycle is a directed vector sum whose axial parts cancel under the exact harmonic structure. The surviving transverse impulse is not produced by assuming transverse instantaneous forces; it is produced by integrating the actual nontransverse waveform.

12. Reproduction procedure

1. Load Q, c_hat, V_E, e0, and A0 in the same ICRS frame.

2. Set m = 1 kg, R = 0.5 m, omega_A = 2 omega_B, omega_B = 1 rad/s, and kappa_F = 1 for the tabulated normalization.

3. For each phase construct e(phi), u(phi), and u_phi.

4. Evaluate B(phi) = u_phi^T Q V_E.

5. Evaluate F_net(phi) = -2 kappa_F m omega_B B(phi)u(phi).

6. Compute F_parallel_scalar = F_net dot c_hat and F_perp = F_net - F_parallel_scalar c_hat.

7. Confirm F_parallel_scalar = -2 kappa_F m omega_B B(phi)sin(2phi).

8. Locate the geometric transverse phases sin(2phi) = 0 and the force-zero roots B(phi) = 0.

9. Integrate F_net with dt = dphi/omega_B over 0 through 2pi.

10. Compare the numerical result with pi kappa_F m c_hat cross(Q V_E).

11. Generate Cell3 vertices, faces, and all 27 nodes from the source basis matrix without changing coordinate frames.

13. Canonical result statement

```
Instantaneous system force:
F_net(phi) = -2 kappa_F m omega_B [u_phi^T Q V_E]u(phi)
Instantaneous CMB-axis component:
F_net(phi) dot c_hat = -2 kappa_F m omega_B [u_phi^T Q V_E]sin(2phi)
Closed-cycle impulse:
J_B = pi kappa_F m c_hat cross(Q V_E)
Therefore: c_hat dot J_B = 0 and c_hat dot <F_net> = 0.
```

The instantaneous net-force vector is not generally perpendicular to the CMB traversal. It becomes purely transverse at the four equatorial phases and purely axial at the four CMB-axis centerline phases. The complete closed-cycle impulse and the cycle-average force are exactly perpendicular to the CMB traversal. The RA = 0 degree seasonal position datum changes the absolute path origin but does not change this directional theorem.

Appendix A. Complete Cell3 boundary-vertex table

#	i	j	k	ICRS x	ICRS y	ICRS z
0	-0.5	-0.5	-0.5	-0.010974902500	0.007385983127	0.039661911012
1	-0.5	-0.5	0.5	-0.011605281985	-0.016871738890	-0.003262443559
2	-0.5	0.5	-0.5	-0.008776156532	0.020288153386	0.005595409476
3	-0.5	0.5	0.5	-0.009406536017	-0.003969568631	-0.037328945095
4	0.5	-0.5	-0.5	0.009406536017	0.003969568631	0.037328945095
5	0.5	-0.5	0.5	0.008776156532	-0.020288153386	-0.005595409476
6	0.5	0.5	-0.5	0.011605281985	0.016871738890	0.003262443559
7	0.5	0.5	0.5	0.010974902500	-0.007385983127	-0.039661911012

Appendix B. Complete 27-node centered half-step registry

#	2i	2j	2k	lambda	ICRS x	ICRS y	ICRS z
0	-1	-1	-1	-3	-0.010974902500	0.007385983127	0.039661911012
1	-1	-1	0	-2	-0.011290092243	-0.004742877881	0.018199733726
2	-1	-1	1	-1	-0.011605281985	-0.016871738890	-0.003262443559
3	-1	0	-1	-2	-0.009875529516	0.013837068257	0.022628660244
4	-1	0	0	-1	-0.010190719258	0.001708207248	0.001166482958
5	-1	0	1	0	-0.010505909001	-0.010420653760	-0.020295694327
6	-1	1	-1	-1	-0.008776156532	0.020288153386	0.005595409476
7	-1	1	0	0	-0.009091346274	0.008159292378	-0.015866767810
8	-1	1	1	1	-0.009406536017	-0.003969568631	-0.037328945095
9	0	-1	-1	-2	-0.000784183241	0.005677775879	0.038495428054
10	0	-1	0	-1	-0.001099372984	-0.006451085129	0.017033250768
11	0	-1	1	0	-0.001414562727	-0.018579946138	-0.004428926518
12	0	0	-1	-1	0.000315189743	0.012128861009	0.021462177286
13	0	0	0	0	0.000000000000	0.000000000000	0.000000000000
14	0	0	1	1	-0.000315189743	-0.012128861009	-0.021462177286
15	0	1	-1	0	0.001414562727	0.018579946138	0.004428926518
16	0	1	0	1	0.001099372984	0.006451085129	-0.017033250768
17	0	1	1	2	0.000784183241	-0.005677775879	-0.038495428054
18	1	-1	-1	-1	0.009406536017	0.003969568631	0.037328945095
19	1	-1	0	0	0.009091346274	-0.008159292378	0.015866767810
20	1	-1	1	1	0.008776156532	-0.020288153386	-0.005595409476
21	1	0	-1	0	0.010505909001	0.010420653760	0.020295694327
22	1	0	0	1	0.010190719258	-0.001708207248	-0.001166482958
23	1	0	1	2	0.009875529516	-0.013837068257	-0.022628660244
24	1	1	-1	1	0.011605281985	0.016871738890	0.003262443559
25	1	1	0	2	0.011290092243	0.004742877881	-0.018199733726
26	1	1	1	3	0.010974902500	-0.007385983127	-0.039661911012

Appendix C. Cell3 face-plane families

Family	Area (native^2)	Included angle (deg)	Unit normal ICRS	Boundary |d|
A_ab	7.547217896726e-04	83.933483012	(0.1940922008, 0.9131771688, 0.3583792348)	0.018828573500
A_ac	1.011255757688e-03	80.448130018	(0.0890525941, 0.8665767693, -0.4910339483)	0.014052166903
A_bc	1.385784431733e-03	50.363231814	(-0.9959651903, 0.0836021625, -0.0326192906)	0.010254361619

Appendix D. Complete one-degree force-direction and cumulative-impulse table

All values use omega_B = 1 rad/s, omega_A = 2 rad/s, kappa_F = 1, m = 1 kg, and R = 0.5 m. The repeated 360-degree row is the closure endpoint. The full vector fields are preserved in the accompanying machine-readable JSON.

Phase	B(phi)	|F| N	F parallel N	|F perp| N	Axis angle deg	Cum J parallel	|Cum J perp|
0	1.46387e+06	2.92773e+06	1.70151e-11	2.92773e+06	90.0000	0.00000e+00	0.00000e+00
1	1.42188e+06	2.84375e+06	-9.92455e+04	2.84202e+06	88.0000	-8.66080e+02	5.03487e+04
2	1.37854e+06	2.75707e+06	-1.92324e+05	2.75036e+06	86.0000	-3.41050e+03	9.91457e+04
3	1.33395e+06	2.66791e+06	-2.78872e+05	2.65329e+06	84.0000	-7.52246e+03	1.46289e+05
4	1.28823e+06	2.57646e+06	-3.58574e+05	2.55139e+06	82.0000	-1.30852e+04	1.91685e+05
5	1.24147e+06	2.48295e+06	-4.31159e+05	2.44523e+06	80.0000	-1.99769e+04	2.35252e+05
6	1.19379e+06	2.38758e+06	-4.96405e+05	2.33540e+06	78.0000	-2.80715e+04	2.76918e+05
7	1.14528e+06	2.29055e+06	-5.54135e+05	2.22251e+06	76.0000	-3.72392e+04	3.16623e+05
8	1.09604e+06	2.19209e+06	-6.04221e+05	2.10717e+06	74.0000	-4.73477e+04	3.54317e+05
9	1.04619e+06	2.09238e+06	-6.46582e+05	1.98997e+06	72.0000	-5.82630e+04	3.89960e+05
10	9.95823e+05	1.99165e+06	-6.81183e+05	1.87153e+06	70.0000	-6.98500e+04	4.23525e+05
11	9.45037e+05	1.89007e+06	-7.08034e+05	1.75245e+06	68.0000	-8.19732e+04	4.54996e+05
12	8.93934e+05	1.78787e+06	-7.27192e+05	1.63330e+06	66.0000	-9.44979e+04	4.84365e+05
13	8.42610e+05	1.68522e+06	-7.38751e+05	1.51466e+06	64.0000	-1.07291e+05	5.11638e+05
14	7.91158e+05	1.58232e+06	-7.42853e+05	1.39710e+06	62.0000	-1.20220e+05	5.36830e+05
15	7.39673e+05	1.47935e+06	-7.39673e+05	1.28115e+06	60.0000	-1.33158e+05	5.59966e+05
16	6.88244e+05	1.37649e+06	-7.29427e+05	1.16733e+06	58.0000	-1.45978e+05	5.81081e+05
17	6.36957e+05	1.27391e+06	-7.12364e+05	1.05612e+06	56.0000	-1.58560e+05	6.00219e+05
18	5.85898e+05	1.17180e+06	-6.88765e+05	9.48003e+05	54.0000	-1.70787e+05	6.17432e+05
19	5.35148e+05	1.07030e+06	-6.58939e+05	8.43404e+05	52.0000	-1.82548e+05	6.32783e+05
20	4.84784e+05	9.69567e+05	-6.23226e+05	7.42731e+05	50.0000	-1.93737e+05	6.46340e+05
21	4.34881e+05	8.69762e+05	-5.81984e+05	6.46359e+05	48.0000	-2.04254e+05	6.58179e+05
22	3.85511e+05	7.71022e+05	-5.35597e+05	5.54626e+05	46.0000	-2.14007e+05	6.68384e+05
23	3.36741e+05	6.73481e+05	-4.84462e+05	4.67839e+05	44.0000	-2.22909e+05	6.77042e+05
24	2.88635e+05	5.77269e+05	-4.28995e+05	3.86268e+05	42.0000	-2.30880e+05	6.84249e+05
25	2.41252e+05	4.82505e+05	-3.69620e+05	3.10148e+05	40.0000	-2.37849e+05	6.90101e+05
26	1.94650e+05	3.89300e+05	-3.06773e+05	2.39677e+05	38.0000	-2.43752e+05	6.94701e+05
27	1.48880e+05	2.97760e+05	-2.40893e+05	1.75019e+05	36.0000	-2.48531e+05	6.98154e+05
28	1.03990e+05	2.07980e+05	-1.72424e+05	1.16301e+05	34.0000	-2.52138e+05	7.00568e+05
29	6.00243e+04	1.20049e+05	-1.01807e+05	6.36161e+04	32.0000	-2.54531e+05	7.02051e+05
30	1.70223e+04	3.40446e+04	-2.94835e+04	1.70223e+04	30.0000	-2.55677e+05	7.02712e+05
31	-2.49802e+04	4.99604e+04	4.41124e+04	2.34550e+04	28.0000	-2.55549e+05	7.02661e+05
32	-6.59517e+04	1.31903e+05	1.18554e+05	5.78227e+04	26.0000	-2.54130e+05	7.02006e+05
33	-1.05865e+05	2.11730e+05	1.93425e+05	8.61183e+04	24.0000	-2.51407e+05	7.00853e+05
34	-1.44697e+05	2.89393e+05	2.68321e+05	1.08409e+05	22.0000	-2.47378e+05	6.99308e+05
35	-1.82428e+05	3.64855e+05	3.42852e+05	1.24788e+05	20.0000	-2.42044e+05	6.97471e+05
36	-2.19043e+05	4.38086e+05	4.16645e+05	1.35376e+05	18.0000	-2.35417e+05	6.95440e+05
37	-2.54532e+05	5.09064e+05	4.89343e+05	1.40317e+05	16.0000	-2.27510e+05	6.93309e+05
38	-2.88887e+05	5.77774e+05	5.60611e+05	1.39776e+05	14.0000	-2.18348e+05	6.91165e+05
39	-3.22105e+05	6.44210e+05	6.30132e+05	1.33939e+05	12.0000	-2.07957e+05	6.89092e+05
40	-3.54186e+05	7.08372e+05	6.97611e+05	1.23008e+05	10.0000	-1.96370e+05	6.87168e+05
41	-3.85135e+05	7.70270e+05	7.62774e+05	1.07201e+05	8.0000	-1.83626e+05	6.85464e+05
42	-4.14959e+05	8.29919e+05	8.25373e+05	8.67501e+04	6.0000	-1.69766e+05	6.84045e+05
43	-4.43670e+05	8.87340e+05	8.85179e+05	6.18977e+04	4.0000	-1.54839e+05	6.82971e+05
44	-4.71282e+05	9.42563e+05	9.41989e+05	3.28950e+04	2.0000	-1.38894e+05	6.82293e+05
45	-4.97811e+05	9.95623e+05	9.95623e+05	1.30156e-10	0.0000	-1.21985e+05	6.82060e+05
46	-5.23280e+05	1.04656e+06	1.04592e+06	3.65244e+04	2.0000	-1.04169e+05	6.82312e+05
47	-5.47710e+05	1.09542e+06	1.09275e+06	7.64127e+04	4.0000	-8.55058e+04	6.83085e+05

Phase	B(phi)	|F| N	F parallel N	|F perp| N	Axis angle deg	Cum J parallel	|Cum J perp|
48	-5.71129e+05	1.14226e+06	1.13600e+06	1.19398e+05	6.0000	-6.60563e+04	6.84409e+05
49	-5.93563e+05	1.18713e+06	1.17557e+06	1.65216e+05	8.0000	-4.58840e+04	6.86309e+05
50	-6.15045e+05	1.23009e+06	1.21140e+06	2.13603e+05	10.0000	-2.50537e+04	6.88807e+05
51	-6.35607e+05	1.27121e+06	1.24344e+06	2.64300e+05	12.0000	-3.63116e+03	6.91920e+05
52	-6.55284e+05	1.31057e+06	1.27164e+06	3.17055e+05	14.0000	1.83170e+04	6.95664e+05
53	-6.74111e+05	1.34822e+06	1.29599e+06	3.71620e+05	16.0000	4.07238e+04	7.00051e+05
54	-6.92128e+05	1.38426e+06	1.31651e+06	4.27758e+05	18.0000	6.35222e+04	7.05089e+05
55	-7.09372e+05	1.41874e+06	1.33318e+06	4.85239e+05	20.0000	8.66451e+04	7.10789e+05
56	-7.25885e+05	1.45177e+06	1.34606e+06	5.43843e+05	22.0000	1.10026e+05	7.17156e+05
57	-7.41708e+05	1.48342e+06	1.35517e+06	6.03360e+05	24.0000	1.33599e+05	7.24197e+05
58	-7.56882e+05	1.51376e+06	1.36056e+06	6.63591e+05	26.0000	1.57298e+05	7.31918e+05
59	-7.71451e+05	1.54290e+06	1.36230e+06	7.24348e+05	28.0000	1.81059e+05	7.40323e+05
60	-7.85455e+05	1.57091e+06	1.36045e+06	7.85455e+05	30.0000	2.04820e+05	7.49417e+05
61	-7.98939e+05	1.59788e+06	1.35508e+06	8.46746e+05	32.0000	2.28517e+05	7.59206e+05
62	-8.11944e+05	1.62389e+06	1.34626e+06	9.08067e+05	34.0000	2.52091e+05	7.69694e+05
63	-8.24513e+05	1.64903e+06	1.33409e+06	9.69273e+05	36.0000	2.75481e+05	7.80885e+05
64	-8.36686e+05	1.67337e+06	1.31864e+06	1.03023e+06	38.0000	2.98631e+05	7.92784e+05
65	-8.48506e+05	1.69701e+06	1.29999e+06	1.09082e+06	40.0000	3.21482e+05	8.05395e+05
66	-8.60010e+05	1.72002e+06	1.27822e+06	1.15092e+06	42.0000	3.43982e+05	8.18721e+05
67	-8.71239e+05	1.74248e+06	1.25343e+06	1.21043e+06	44.0000	3.66074e+05	8.32766e+05
68	-8.82229e+05	1.76446e+06	1.22570e+06	1.26924e+06	46.0000	3.87709e+05	8.47532e+05
69	-8.93016e+05	1.78603e+06	1.19509e+06	1.32728e+06	48.0000	4.08834e+05	8.63022e+05
70	-9.03633e+05	1.80727e+06	1.16169e+06	1.38445e+06	50.0000	4.29401e+05	8.79235e+05
71	-9.14114e+05	1.82823e+06	1.12557e+06	1.44066e+06	52.0000	4.49361e+05	8.96172e+05
72	-9.24487e+05	1.84897e+06	1.08680e+06	1.49585e+06	54.0000	4.68668e+05	9.13830e+05
73	-9.34781e+05	1.86956e+06	1.04545e+06	1.54994e+06	56.0000	4.87275e+05	9.32208e+05
74	-9.45022e+05	1.89004e+06	1.00157e+06	1.60285e+06	58.0000	5.05139e+05	9.51300e+05
75	-9.55232e+05	1.91046e+06	9.55232e+05	1.65451e+06	60.0000	5.22215e+05	9.71101e+05
76	-9.65432e+05	1.93086e+06	9.06486e+05	1.70485e+06	62.0000	5.38461e+05	9.91603e+05
77	-9.75640e+05	1.95128e+06	8.55385e+05	1.75380e+06	64.0000	5.53837e+05	1.01280e+06
78	-9.85872e+05	1.97174e+06	8.01980e+05	1.80128e+06	66.0000	5.68300e+05	1.03467e+06
79	-9.96138e+05	1.99228e+06	7.46320e+05	1.84721e+06	68.0000	5.81811e+05	1.05721e+06
80	-1.00645e+06	2.01290e+06	6.88452e+05	1.89151e+06	70.0000	5.94332e+05	1.08039e+06
81	-1.01681e+06	2.03362e+06	6.28424e+05	1.93409e+06	72.0000	6.05824e+05	1.10421e+06
82	-1.02723e+06	2.05446e+06	5.66285e+05	1.97487e+06	74.0000	6.16250e+05	1.12865e+06
83	-1.03770e+06	2.07540e+06	5.02084e+05	2.01375e+06	76.0000	6.25573e+05	1.15367e+06
84	-1.04822e+06	2.09644e+06	4.35875e+05	2.05063e+06	78.0000	6.33758e+05	1.17926e+06
85	-1.05879e+06	2.11758e+06	3.67713e+05	2.08541e+06	80.0000	6.40771e+05	1.20539e+06
86	-1.06939e+06	2.13878e+06	2.97661e+05	2.11797e+06	82.0000	6.46578e+05	1.23203e+06
87	-1.08002e+06	2.16003e+06	2.25785e+05	2.14820e+06	84.0000	6.51145e+05	1.25915e+06
88	-1.09065e+06	2.18130e+06	1.52160e+05	2.17599e+06	86.0000	6.54444e+05	1.28671e+06
89	-1.10127e+06	2.20254e+06	7.68677e+04	2.20120e+06	88.0000	6.56442e+05	1.31467e+06
90	-1.11186e+06	2.22372e+06	2.71978e-10	2.22372e+06	90.0000	6.57113e+05	1.34301e+06
91	-1.12239e+06	2.24478e+06	-7.83418e+04	2.24341e+06	88.0000	6.56429e+05	1.37168e+06
92	-1.13284e+06	2.26567e+06	-1.58045e+05	2.26015e+06	86.0000	6.54367e+05	1.40062e+06
93	-1.14316e+06	2.28633e+06	-2.38986e+05	2.27380e+06	84.0000	6.50902e+05	1.42981e+06
94	-1.15334e+06	2.30668e+06	-3.21028e+05	2.28424e+06	82.0000	6.46015e+05	1.45918e+06
95	-1.16333e+06	2.32667e+06	-4.04021e+05	2.29132e+06	80.0000	6.39688e+05	1.48870e+06
96	-1.17310e+06	2.34620e+06	-4.87802e+05	2.29493e+06	78.0000	6.31905e+05	1.51829e+06
97	-1.18260e+06	2.36520e+06	-5.72193e+05	2.29494e+06	76.0000	6.22655e+05	1.54793e+06
98	-1.19179e+06	2.38358e+06	-6.57003e+05	2.29124e+06	74.0000	6.11928e+05	1.57753e+06
99	-1.20062e+06	2.40125e+06	-7.42026e+05	2.28372e+06	72.0000	5.99719e+05	1.60705e+06
100	-1.20905e+06	2.41811e+06	-8.27042e+05	2.27228e+06	70.0000	5.86026e+05	1.63643e+06
101	-1.21703e+06	2.43406e+06	-9.11816e+05	2.25682e+06	68.0000	5.70852e+05	1.66561e+06

Phase	B(phi)	|F| N	F parallel N	|F perp| N	Axis angle deg	Cum J parallel	|Cum J perp|
102	-1.22450e+06	2.44901e+06	-9.96102e+05	2.23728e+06	66.0000	5.54202e+05	1.69452e+06
103	-1.23142e+06	2.46284e+06	-1.07964e+06	2.21359e+06	64.0000	5.36088e+05	1.72311e+06
104	-1.23772e+06	2.47545e+06	-1.16215e+06	2.18569e+06	62.0000	5.16525e+05	1.75130e+06
105	-1.24336e+06	2.48672e+06	-1.24336e+06	2.15356e+06	60.0000	4.95533e+05	1.77905e+06
106	-1.24827e+06	2.49655e+06	-1.32297e+06	2.11719e+06	58.0000	4.73137e+05	1.80627e+06
107	-1.25241e+06	2.50482e+06	-1.40068e+06	2.07659e+06	56.0000	4.49369e+05	1.83292e+06
108	-1.25570e+06	2.51141e+06	-1.47617e+06	2.03177e+06	54.0000	4.24264e+05	1.85893e+06
109	-1.25810e+06	2.51620e+06	-1.54913e+06	1.98280e+06	52.0000	3.97863e+05	1.88425e+06
110	-1.25955e+06	2.51910e+06	-1.61924e+06	1.92974e+06	50.0000	3.70214e+05	1.90881e+06
111	-1.25998e+06	2.51996e+06	-1.68618e+06	1.87270e+06	48.0000	3.41368e+05	1.93256e+06
112	-1.25935e+06	2.51869e+06	-1.74963e+06	1.81180e+06	46.0000	3.11385e+05	1.95544e+06
113	-1.25758e+06	2.51517e+06	-1.80926e+06	1.74718e+06	44.0000	2.80328e+05	1.97742e+06
114	-1.25464e+06	2.50928e+06	-1.86476e+06	1.67904e+06	42.0000	2.48266e+05	1.99843e+06
115	-1.25046e+06	2.50092e+06	-1.91581e+06	1.60756e+06	40.0000	2.15275e+05	2.01845e+06
116	-1.24498e+06	2.48997e+06	-1.96212e+06	1.53298e+06	38.0000	1.81433e+05	2.03743e+06
117	-1.23817e+06	2.47633e+06	-2.00339e+06	1.45555e+06	36.0000	1.46827e+05	2.05534e+06
118	-1.22995e+06	2.45991e+06	-2.03935e+06	1.37556e+06	34.0000	1.11548e+05	2.07216e+06
119	-1.22030e+06	2.44059e+06	-2.06974e+06	1.29332e+06	32.0000	7.56892e+04	2.08786e+06
120	-1.20915e+06	2.41830e+06	-2.09431e+06	1.20915e+06	30.0000	3.93510e+04	2.10243e+06
121	-1.19647e+06	2.39294e+06	-2.11284e+06	1.12342e+06	28.0000	2.63678e+03	2.11586e+06
122	-1.18221e+06	2.36442e+06	-2.12513e+06	1.03649e+06	26.0000	-3.43464e+04	2.12815e+06
123	-1.16634e+06	2.33267e+06	-2.13100e+06	9.48782e+05	24.0000	-7.14881e+04	2.13929e+06
124	-1.14881e+06	2.29762e+06	-2.13031e+06	8.60703e+05	22.0000	-1.08675e+05	2.14932e+06
125	-1.12960e+06	2.25920e+06	-2.12295e+06	7.72691e+05	20.0000	-1.45792e+05	2.15823e+06
126	-1.10867e+06	2.21735e+06	-2.10882e+06	6.85198e+05	18.0000	-1.82721e+05	2.16606e+06
127	-1.08601e+06	2.17202e+06	-2.08788e+06	5.98690e+05	16.0000	-2.19344e+05	2.17284e+06
128	-1.06158e+06	2.12316e+06	-2.06010e+06	5.13640e+05	14.0000	-2.55542e+05	2.17861e+06
129	-1.03537e+06	2.07075e+06	-2.02550e+06	4.30533e+05	12.0000	-2.91196e+05	2.18342e+06
130	-1.00737e+06	2.01474e+06	-1.98413e+06	3.49856e+05	10.0000	-3.26186e+05	2.18731e+06
131	-9.77560e+05	1.95512e+06	-1.93609e+06	2.72100e+05	8.0000	-3.60397e+05	2.19034e+06
132	-9.45938e+05	1.89188e+06	-1.88151e+06	1.97755e+05	6.0000	-3.93712e+05	2.19258e+06
133	-9.12503e+05	1.82501e+06	-1.82056e+06	1.27306e+05	4.0000	-4.26018e+05	2.19410e+06
134	-8.77257e+05	1.75451e+06	-1.75345e+06	6.12317e+04	2.0000	-4.57207e+05	2.19495e+06
135	-8.40208e+05	1.68042e+06	-1.68042e+06	3.40652e-10	0.0000	-4.87173e+05	2.19522e+06
136	-8.01367e+05	1.60273e+06	-1.60176e+06	5.59346e+04	2.0000	-5.15816e+05	2.19499e+06
137	-7.60753e+05	1.52151e+06	-1.51780e+06	1.06135e+05	4.0000	-5.43039e+05	2.19433e+06
138	-7.18385e+05	1.43677e+06	-1.42890e+06	1.50183e+05	6.0000	-5.68754e+05	2.19331e+06
139	-6.74292e+05	1.34858e+06	-1.33546e+06	1.87686e+05	8.0000	-5.92877e+05	2.19202e+06
140	-6.28503e+05	1.25701e+06	-1.23791e+06	2.18277e+05	10.0000	-6.15334e+05	2.19052e+06
141	-5.81057e+05	1.16211e+06	-1.13672e+06	2.41617e+05	12.0000	-6.36057e+05	2.18890e+06
142	-5.31993e+05	1.06399e+06	-1.03238e+06	2.57402e+05	14.0000	-6.54986e+05	2.18721e+06
143	-4.81358e+05	9.62717e+05	-9.25423e+05	2.65361e+05	16.0000	-6.72071e+05	2.18553e+06
144	-4.29204e+05	8.58407e+05	-8.16394e+05	2.65262e+05	18.0000	-6.87271e+05	2.18390e+06
145	-3.75585e+05	7.51170e+05	-7.05869e+05	2.56915e+05	20.0000	-7.00555e+05	2.18239e+06
146	-3.20563e+05	6.41125e+05	-5.94441e+05	2.40170e+05	22.0000	-7.11903e+05	2.18102e+06
147	-2.64202e+05	5.28405e+05	-4.82722e+05	2.14922e+05	24.0000	-7.21303e+05	2.17984e+06
148	-2.06574e+05	4.13148e+05	-3.71335e+05	1.81112e+05	26.0000	-7.28756e+05	2.17888e+06
149	-1.47752e+05	2.95503e+05	-2.60914e+05	1.38730e+05	28.0000	-7.34273e+05	2.17815e+06
150	-8.78148e+04	1.75630e+05	-1.52100e+05	8.78148e+04	30.0000	-7.37877e+05	2.17767e+06
151	-2.68466e+04	5.36931e+04	-4.55343e+04	2.84530e+04	32.0000	-7.39602e+05	2.17743e+06
152	3.50658e+04	7.01315e+04	5.81417e+04	3.92170e+04	34.0000	-7.39492e+05	2.17745e+06
153	9.78306e+04	1.95661e+05	1.58293e+05	1.15007e+05	36.0000	-7.37603e+05	2.17770e+06
154	1.61353e+05	3.22706e+05	2.54295e+05	1.98677e+05	38.0000	-7.34003e+05	2.17817e+06
155	2.25533e+05	4.51067e+05	3.45537e+05	2.89940e+05	40.0000	-7.28768e+05	2.17884e+06

Phase	B(phi)	|F| N	F parallel N	|F perp| N	Axis angle deg	Cum J parallel	|Cum J perp|
156	2.90269e+05	5.80539e+05	4.31425e+05	3.88456e+05	42.0000	-7.21988e+05	2.17968e+06
157	3.55456e+05	7.10912e+05	5.11387e+05	4.93841e+05	44.0000	-7.13760e+05	2.18067e+06
158	4.20983e+05	8.41966e+05	5.84879e+05	6.05660e+05	46.0000	-7.04194e+05	2.18177e+06
159	4.86740e+05	9.73480e+05	6.51385e+05	7.23437e+05	48.0000	-6.93405e+05	2.18296e+06
160	5.52612e+05	1.10522e+06	7.10424e+05	8.46650e+05	50.0000	-6.81521e+05	2.18421e+06
161	6.18482e+05	1.23696e+06	7.61551e+05	9.74741e+05	52.0000	-6.68676e+05	2.18548e+06
162	6.84231e+05	1.36846e+06	8.04362e+05	1.10711e+06	54.0000	-6.55011e+05	2.18677e+06
163	7.49739e+05	1.49948e+06	8.38498e+05	1.24312e+06	56.0000	-6.40674e+05	2.18804e+06
164	8.14884e+05	1.62977e+06	8.63645e+05	1.38212e+06	58.0000	-6.25820e+05	2.18930e+06
165	8.79540e+05	1.75908e+06	8.79540e+05	1.52341e+06	60.0000	-6.10608e+05	2.19052e+06
166	9.43584e+05	1.88717e+06	8.85972e+05	1.66627e+06	62.0000	-5.95201e+05	2.19172e+06
167	1.00689e+06	2.01378e+06	8.82783e+05	1.80997e+06	64.0000	-5.79766e+05	2.19290e+06
168	1.06933e+06	2.13866e+06	8.69873e+05	1.95377e+06	66.0000	-5.64471e+05	2.19408e+06
169	1.13078e+06	2.26157e+06	8.47198e+05	2.09689e+06	68.0000	-5.49486e+05	2.19528e+06
170	1.19112e+06	2.38224e+06	8.14773e+05	2.23857e+06	70.0000	-5.34983e+05	2.19653e+06
171	1.25021e+06	2.50042e+06	7.72672e+05	2.37804e+06	72.0000	-5.21130e+05	2.19788e+06
172	1.30793e+06	2.61587e+06	7.21031e+05	2.51454e+06	74.0000	-5.08095e+05	2.19937e+06
173	1.36417e+06	2.72834e+06	6.60044e+05	2.64729e+06	76.0000	-4.96043e+05	2.20106e+06
174	1.41879e+06	2.83757e+06	5.89965e+05	2.77557e+06	78.0000	-4.85134e+05	2.20300e+06
175	1.47167e+06	2.94334e+06	5.11106e+05	2.89863e+06	80.0000	-4.75526e+05	2.20525e+06
176	1.52270e+06	3.04540e+06	4.23838e+05	3.01576e+06	82.0000	-4.67367e+05	2.20788e+06
177	1.57176e+06	3.14352e+06	3.28587e+05	3.12630e+06	84.0000	-4.60801e+05	2.21095e+06
178	1.61873e+06	3.23746e+06	2.25834e+05	3.22958e+06	86.0000	-4.55962e+05	2.21452e+06
179	1.66351e+06	3.32701e+06	1.16111e+05	3.32499e+06	88.0000	-4.52978e+05	2.21866e+06
180	1.70598e+06	3.41195e+06	1.03126e-09	3.41195e+06	90.0000	-4.51965e+05	2.22342e+06
181	1.74604e+06	3.49207e+06	-1.21872e+05	3.48995e+06	88.0000	-4.53029e+05	2.22885e+06
182	1.78359e+06	3.56717e+06	-2.48833e+05	3.55848e+06	86.0000	-4.56264e+05	2.23500e+06
183	1.81853e+06	3.63705e+06	-3.80175e+05	3.61713e+06	84.0000	-4.61753e+05	2.24188e+06
184	1.85076e+06	3.70153e+06	-5.15153e+05	3.66551e+06	82.0000	-4.69566e+05	2.24953e+06
185	1.88021e+06	3.76043e+06	-6.52991e+05	3.70330e+06	80.0000	-4.79760e+05	2.25796e+06
186	1.90679e+06	3.81357e+06	-7.92887e+05	3.73024e+06	78.0000	-4.92378e+05	2.26717e+06
187	1.93041e+06	3.86082e+06	-9.34017e+05	3.74614e+06	76.0000	-5.07448e+05	2.27713e+06
188	1.95101e+06	3.90201e+06	-1.07554e+06	3.75085e+06	74.0000	-5.24984e+05	2.28782e+06
189	1.96851e+06	3.93702e+06	-1.21661e+06	3.74433e+06	72.0000	-5.44987e+05	2.29921e+06
190	1.98286e+06	3.96571e+06	-1.35635e+06	3.72655e+06	70.0000	-5.67440e+05	2.31122e+06
191	1.99399e+06	3.98798e+06	-1.49392e+06	3.69759e+06	68.0000	-5.92314e+05	2.32382e+06
192	2.00186e+06	4.00372e+06	-1.62846e+06	3.65758e+06	66.0000	-6.19562e+05	2.33690e+06
193	2.00643e+06	4.01285e+06	-1.75912e+06	3.60673e+06	64.0000	-6.49124e+05	2.35040e+06
194	2.00765e+06	4.01529e+06	-1.88507e+06	3.54529e+06	62.0000	-6.80926e+05	2.36422e+06
195	2.00549e+06	4.01098e+06	-2.00549e+06	3.47361e+06	60.0000	-7.14877e+05	2.37827e+06
196	1.99994e+06	3.99987e+06	-2.11961e+06	3.39208e+06	58.0000	-7.50875e+05	2.39243e+06
197	1.99096e+06	3.98192e+06	-2.22666e+06	3.30116e+06	56.0000	-7.88804e+05	2.40661e+06
198	1.97856e+06	3.95712e+06	-2.32594e+06	3.20138e+06	54.0000	-8.28533e+05	2.42071e+06
199	1.96272e+06	3.92545e+06	-2.41675e+06	3.09329e+06	52.0000	-8.69920e+05	2.43461e+06
200	1.94346e+06	3.88691e+06	-2.49846e+06	2.97755e+06	50.0000	-9.12814e+05	2.44822e+06
201	1.92077e+06	3.84154e+06	-2.57049e+06	2.85482e+06	48.0000	-9.57049e+05	2.46145e+06
202	1.89468e+06	3.78937e+06	-2.63232e+06	2.72584e+06	46.0000	-1.00245e+06	2.47421e+06
203	1.86522e+06	3.73044e+06	-2.68345e+06	2.59138e+06	44.0000	-1.04884e+06	2.48641e+06
204	1.83240e+06	3.66481e+06	-2.72348e+06	2.45224e+06	42.0000	-1.09602e+06	2.49799e+06
205	1.79628e+06	3.59257e+06	-2.75207e+06	2.30926e+06	40.0000	-1.14381e+06	2.50889e+06
206	1.75690e+06	3.51380e+06	-2.76891e+06	2.16331e+06	38.0000	-1.19199e+06	2.51905e+06
207	1.71431e+06	3.42861e+06	-2.77381e+06	2.01529e+06	36.0000	-1.24036e+06	2.52844e+06
208	1.66856e+06	3.33713e+06	-2.76660e+06	1.86610e+06	34.0000	-1.28871e+06	2.53703e+06
209	1.61974e+06	3.23947e+06	-2.74723e+06	1.71666e+06	32.0000	-1.33682e+06	2.54481e+06

Phase	B(phi)	|F| N	F parallel N	|F perp| N	Axis angle deg	Cum J parallel	|Cum J perp|
210	1.56790e+06	3.13580e+06	-2.71568e+06	1.56790e+06	30.0000	-1.38450e+06	2.55178e+06
211	1.51313e+06	3.02626e+06	-2.67203e+06	1.42074e+06	28.0000	-1.43151e+06	2.55794e+06
212	1.45552e+06	2.91104e+06	-2.61642e+06	1.27612e+06	26.0000	-1.47766e+06	2.56332e+06
213	1.39516e+06	2.79031e+06	-2.54908e+06	1.13492e+06	24.0000	-1.52274e+06	2.56794e+06
214	1.33214e+06	2.66428e+06	-2.47028e+06	9.98057e+05	22.0000	-1.56654e+06	2.57185e+06
215	1.26658e+06	2.53316e+06	-2.38039e+06	8.66390e+05	20.0000	-1.60887e+06	2.57510e+06
216	1.19858e+06	2.39716e+06	-2.27983e+06	7.40762e+05	18.0000	-1.64954e+06	2.57774e+06
217	1.12826e+06	2.25652e+06	-2.16910e+06	6.21981e+05	16.0000	-1.68837e+06	2.57983e+06
218	1.05574e+06	2.11148e+06	-2.04876e+06	5.10814e+05	14.0000	-1.72517e+06	2.58144e+06
219	9.81152e+05	1.96230e+06	-1.91942e+06	4.07986e+05	12.0000	-1.75980e+06	2.58264e+06
220	9.04624e+05	1.80925e+06	-1.78176e+06	3.14173e+05	10.0000	-1.79210e+06	2.58349e+06
221	8.26292e+05	1.65258e+06	-1.63650e+06	2.29995e+05	8.0000	-1.82193e+06	2.58405e+06
222	7.46298e+05	1.49260e+06	-1.48442e+06	1.56019e+05	6.0000	-1.84917e+06	2.58441e+06
223	6.64787e+05	1.32957e+06	-1.32634e+06	9.27464e+04	4.0000	-1.87370e+06	2.58460e+06
224	5.81908e+05	1.16382e+06	-1.16311e+06	4.06166e+04	2.0000	-1.89542e+06	2.58468e+06
225	4.97811e+05	9.95623e+05	-9.95623e+05	3.33426e-10	0.0000	-1.91426e+06	2.58470e+06
226	4.12654e+05	8.25308e+05	-8.24805e+05	2.88028e+04	2.0000	-1.93014e+06	2.58470e+06
227	3.26593e+05	6.53186e+05	-6.51595e+05	4.55640e+04	4.0000	-1.94303e+06	2.58469e+06
228	2.39790e+05	4.79580e+05	-4.76952e+05	5.01297e+04	6.0000	-1.95288e+06	2.58469e+06
229	1.52407e+05	3.04813e+05	-3.01847e+05	4.24218e+04	8.0000	-1.95967e+06	2.58470e+06
230	6.46079e+04	1.29216e+05	-1.27253e+05	2.24381e+04	10.0000	-1.96342e+06	2.58472e+06
231	-2.34402e+04	4.68804e+04	4.58560e+04	9.74699e+03	12.0000	-1.96413e+06	2.58472e+06
232	-1.11571e+05	2.23142e+05	2.16513e+05	5.39828e+04	14.0000	-1.96184e+06	2.58468e+06
233	-1.99616e+05	3.99232e+05	3.83766e+05	1.10043e+05	16.0000	-1.95660e+06	2.58455e+06
234	-2.87408e+05	5.74816e+05	5.46682e+05	1.77628e+05	18.0000	-1.94848e+06	2.58429e+06
235	-3.74778e+05	7.49556e+05	7.04352e+05	2.56363e+05	20.0000	-1.93756e+06	2.58382e+06
236	-4.61559e+05	9.23118e+05	8.55900e+05	3.45806e+05	22.0000	-1.92395e+06	2.58310e+06
237	-5.47583e+05	1.09517e+06	1.00048e+06	4.45444e+05	24.0000	-1.90775e+06	2.58205e+06
238	-6.32685e+05	1.26537e+06	1.13731e+06	5.54702e+05	26.0000	-1.88909e+06	2.58060e+06
239	-7.16701e+05	1.43340e+06	1.26562e+06	6.72941e+05	28.0000	-1.86812e+06	2.57868e+06
240	-7.99466e+05	1.59893e+06	1.38472e+06	7.99466e+05	30.0000	-1.84499e+06	2.57621e+06
241	-8.80822e+05	1.76164e+06	1.49396e+06	9.33529e+05	32.0000	-1.81987e+06	2.57312e+06
242	-9.60610e+05	1.92122e+06	1.59276e+06	1.07433e+06	34.0000	-1.79294e+06	2.56935e+06
243	-1.03867e+06	2.07735e+06	1.68061e+06	1.22104e+06	36.0000	-1.76437e+06	2.56484e+06
244	-1.11486e+06	2.22973e+06	1.75705e+06	1.37276e+06	38.0000	-1.73437e+06	2.55954e+06
245	-1.18903e+06	2.37806e+06	1.82170e+06	1.52859e+06	40.0000	-1.70314e+06	2.55341e+06
246	-1.26103e+06	2.52206e+06	1.87425e+06	1.68759e+06	42.0000	-1.67089e+06	2.54641e+06
247	-1.33072e+06	2.66144e+06	1.91448e+06	1.84879e+06	44.0000	-1.63782e+06	2.53854e+06
248	-1.39797e+06	2.79593e+06	1.94222e+06	2.01122e+06	46.0000	-1.60417e+06	2.52977e+06
249	-1.46264e+06	2.92527e+06	1.95739e+06	2.17390e+06	48.0000	-1.57014e+06	2.52012e+06
250	-1.52461e+06	3.04922e+06	1.96000e+06	2.33583e+06	50.0000	-1.53595e+06	2.50961e+06
251	-1.58376e+06	3.16751e+06	1.95012e+06	2.49604e+06	52.0000	-1.50183e+06	2.49829e+06
252	-1.63997e+06	3.27994e+06	1.92790e+06	2.65353e+06	54.0000	-1.46799e+06	2.48618e+06
253	-1.69314e+06	3.38628e+06	1.89358e+06	2.80735e+06	56.0000	-1.43464e+06	2.47337e+06
254	-1.74316e+06	3.48632e+06	1.84747e+06	2.95656e+06	58.0000	-1.40199e+06	2.45993e+06
255	-1.78993e+06	3.57987e+06	1.78993e+06	3.10026e+06	60.0000	-1.37025e+06	2.44595e+06
256	-1.83337e+06	3.66675e+06	1.72143e+06	3.23755e+06	62.0000	-1.33961e+06	2.43153e+06
257	-1.87340e+06	3.74679e+06	1.64249e+06	3.36759e+06	64.0000	-1.31025e+06	2.41678e+06
258	-1.90992e+06	3.81985e+06	1.55367e+06	3.48961e+06	66.0000	-1.28236e+06	2.40181e+06
259	-1.94289e+06	3.88578e+06	1.45564e+06	3.60283e+06	68.0000	-1.25610e+06	2.38676e+06
260	-1.97223e+06	3.94446e+06	1.34908e+06	3.70658e+06	70.0000	-1.23162e+06	2.37175e+06
261	-1.99789e+06	3.99578e+06	1.23476e+06	3.80021e+06	72.0000	-1.20907e+06	2.35692e+06
262	-2.01982e+06	4.03964e+06	1.11348e+06	3.88315e+06	74.0000	-1.18858e+06	2.34239e+06
263	-2.03799e+06	4.07597e+06	9.86067e+05	3.95490e+06	76.0000	-1.17026e+06	2.32829e+06

Phase	B(phi)	|F| N	F parallel N	|F perp| N	Axis angle deg	Cum J parallel	|Cum J perp|
264	-2.05235e+06	4.10471e+06	8.53417e+05	4.01501e+06	78.0000	-1.15421e+06	2.31475e+06
265	-2.06290e+06	4.12580e+06	7.16437e+05	4.06312e+06	80.0000	-1.14051e+06	2.30186e+06
266	-2.06960e+06	4.13921e+06	5.76066e+05	4.09893e+06	82.0000	-1.12923e+06	2.28975e+06
267	-2.07246e+06	4.14492e+06	4.33263e+05	4.12222e+06	84.0000	-1.12042e+06	2.27849e+06
268	-2.07147e+06	4.14294e+06	2.88997e+05	4.13285e+06	86.0000	-1.11412e+06	2.26817e+06
269	-2.06664e+06	4.13328e+06	1.44249e+05	4.13076e+06	88.0000	-1.11034e+06	2.25884e+06
270	-2.05798e+06	4.11597e+06	1.56465e-09	4.11597e+06	90.0000	-1.10908e+06	2.25054e+06
271	-2.04552e+06	4.09104e+06	-1.42775e+05	4.08855e+06	88.0000	-1.11032e+06	2.24330e+06
272	-2.02929e+06	4.05857e+06	-2.83112e+05	4.04869e+06	86.0000	-1.11404e+06	2.23713e+06
273	-2.00931e+06	4.01863e+06	-4.20061e+05	3.99662e+06	84.0000	-1.12018e+06	2.23200e+06
274	-1.98565e+06	3.97131e+06	-5.52699e+05	3.93266e+06	82.0000	-1.12867e+06	2.22790e+06
275	-1.95835e+06	3.91671e+06	-6.80129e+05	3.85720e+06	80.0000	-1.13942e+06	2.22477e+06
276	-1.92748e+06	3.85495e+06	-8.01489e+05	3.77071e+06	78.0000	-1.15235e+06	2.22256e+06
277	-1.89309e+06	3.78617e+06	-9.15958e+05	3.67371e+06	76.0000	-1.16734e+06	2.22118e+06
278	-1.85526e+06	3.71052e+06	-1.02276e+06	3.56678e+06	74.0000	-1.18426e+06	2.22057e+06
279	-1.81408e+06	3.62815e+06	-1.12116e+06	3.45058e+06	72.0000	-1.20297e+06	2.22062e+06
280	-1.76962e+06	3.53925e+06	-1.21049e+06	3.32581e+06	70.0000	-1.22332e+06	2.22124e+06
281	-1.72200e+06	3.44399e+06	-1.29014e+06	3.19321e+06	68.0000	-1.24514e+06	2.22233e+06
282	-1.67129e+06	3.34258e+06	-1.35955e+06	3.05360e+06	66.0000	-1.26826e+06	2.22380e+06
283	-1.61762e+06	3.23523e+06	-1.41823e+06	2.90781e+06	64.0000	-1.29250e+06	2.22553e+06
284	-1.56108e+06	3.12216e+06	-1.46577e+06	2.75671e+06	62.0000	-1.31767e+06	2.22744e+06
285	-1.50180e+06	3.00361e+06	-1.50180e+06	2.60120e+06	60.0000	-1.34357e+06	2.22944e+06
286	-1.43991e+06	2.87981e+06	-1.52607e+06	2.44222e+06	58.0000	-1.36999e+06	2.23145e+06
287	-1.37551e+06	2.75102e+06	-1.53835e+06	2.28070e+06	56.0000	-1.39673e+06	2.23340e+06
288	-1.30875e+06	2.61751e+06	-1.53853e+06	2.11761e+06	54.0000	-1.42358e+06	2.23523e+06
289	-1.23977e+06	2.47954e+06	-1.52656e+06	1.95390e+06	52.0000	-1.45033e+06	2.23690e+06
290	-1.16869e+06	2.33739e+06	-1.50244e+06	1.79054e+06	50.0000	-1.47676e+06	2.23835e+06
291	-1.09567e+06	2.19134e+06	-1.46629e+06	1.62848e+06	48.0000	-1.50267e+06	2.23958e+06
292	-1.02085e+06	2.04170e+06	-1.41828e+06	1.46867e+06	46.0000	-1.52784e+06	2.24056e+06
293	-9.44374e+05	1.88875e+06	-1.35865e+06	1.31203e+06	44.0000	-1.55208e+06	2.24130e+06
294	-8.66399e+05	1.73280e+06	-1.28772e+06	1.15947e+06	42.0000	-1.57517e+06	2.24180e+06
295	-7.87078e+05	1.57416e+06	-1.20587e+06	1.01185e+06	40.0000	-1.59693e+06	2.24209e+06
296	-7.06567e+05	1.41313e+06	-1.11356e+06	8.70012e+05	38.0000	-1.61717e+06	2.24218e+06
297	-6.25021e+05	1.25004e+06	-1.01131e+06	7.34757e+05	36.0000	-1.63572e+06	2.24210e+06
298	-5.42601e+05	1.08520e+06	-8.99673e+05	6.06837e+05	34.0000	-1.65239e+06	2.24191e+06
299	-4.59465e+05	9.18929e+05	-7.79296e+05	4.86958e+05	32.0000	-1.66704e+06	2.24163e+06
300	-3.75772e+05	7.51544e+05	-6.50857e+05	3.75772e+05	30.0000	-1.67952e+06	2.24130e+06
301	-2.91684e+05	5.83367e+05	-5.15083e+05	2.73874e+05	28.0000	-1.68970e+06	2.24098e+06
302	-2.07359e+05	4.14717e+05	-3.72746e+05	1.81800e+05	26.0000	-1.69745e+06	2.24069e+06
303	-1.22956e+05	2.45913e+05	-2.24653e+05	1.00022e+05	24.0000	-1.70266e+06	2.24048e+06
304	-3.86350e+04	7.72700e+04	-7.16435e+04	2.89459e+04	22.0000	-1.70525e+06	2.24036e+06
305	4.54485e+04	9.08970e+04	8.54153e+04	3.10886e+04	20.0000	-1.70513e+06	2.24037e+06
306	1.29139e+05	2.58277e+05	2.45636e+05	7.98121e+04	18.0000	-1.70224e+06	2.24051e+06
307	2.12282e+05	4.24564e+05	4.08117e+05	1.17026e+05	16.0000	-1.69653e+06	2.24079e+06
308	2.94728e+05	5.89455e+05	5.71946e+05	1.42602e+05	14.0000	-1.68798e+06	2.24119e+06
309	3.76326e+05	7.52652e+05	7.36205e+05	1.56485e+05	12.0000	-1.67656e+06	2.24170e+06
310	4.56932e+05	9.13865e+05	8.99981e+05	1.58691e+05	10.0000	-1.66229e+06	2.24229e+06
311	5.36403e+05	1.07281e+06	1.06237e+06	1.49306e+05	8.0000	-1.64516e+06	2.24291e+06
312	6.14600e+05	1.22920e+06	1.22247e+06	1.28486e+05	6.0000	-1.62522e+06	2.24351e+06
313	6.91386e+05	1.38277e+06	1.37940e+06	9.64573e+04	4.0000	-1.60252e+06	2.24403e+06
314	7.66631e+05	1.53326e+06	1.53233e+06	5.35101e+04	2.0000	-1.57711e+06	2.24440e+06
315	8.40208e+05	1.68042e+06	1.68042e+06	8.43509e-10	0.0000	-1.54907e+06	2.24454e+06
316	9.11993e+05	1.82399e+06	1.82288e+06	6.36562e+04	2.0000	-1.51850e+06	2.24436e+06
317	9.81870e+05	1.96374e+06	1.95896e+06	1.36984e+05	4.0000	-1.48549e+06	2.24377e+06

Phase	B(phi)	|F| N	F parallel N	|F perp| N	Axis angle deg	Cum J parallel	|Cum J perp|
318	1.04972e+06	2.09945e+06	2.08795e+06	2.19452e+05	6.0000	-1.45018e+06	2.24267e+06
319	1.11545e+06	2.23090e+06	2.20919e+06	3.10481e+05	8.0000	-1.41268e+06	2.24098e+06
320	1.17894e+06	2.35788e+06	2.32206e+06	4.09442e+05	10.0000	-1.37314e+06	2.23859e+06
321	1.24010e+06	2.48021e+06	2.42601e+06	5.15664e+05	12.0000	-1.33170e+06	2.23542e+06
322	1.29885e+06	2.59770e+06	2.52053e+06	6.28439e+05	14.0000	-1.28854e+06	2.23137e+06
323	1.35509e+06	2.71017e+06	2.60518e+06	7.47024e+05	16.0000	-1.24381e+06	2.22637e+06
324	1.40874e+06	2.81748e+06	2.67958e+06	8.70649e+05	18.0000	-1.19769e+06	2.22032e+06
325	1.45974e+06	2.91947e+06	2.74340e+06	9.98518e+05	20.0000	-1.15036e+06	2.21318e+06
326	1.50801e+06	3.01601e+06	2.79640e+06	1.12982e+06	22.0000	-1.10202e+06	2.20487e+06
327	1.55349e+06	3.10699e+06	2.83837e+06	1.26373e+06	24.0000	-1.05285e+06	2.19534e+06
328	1.59614e+06	3.19228e+06	2.86921e+06	1.39940e+06	26.0000	-1.00304e+06	2.18458e+06
329	1.63590e+06	3.27181e+06	2.88883e+06	1.53602e+06	28.0000	-9.52789e+05	2.17253e+06
330	1.67274e+06	3.34547e+06	2.89726e+06	1.67274e+06	30.0000	-9.02296e+05	2.15921e+06
331	1.70661e+06	3.41322e+06	2.89457e+06	1.80873e+06	32.0000	-8.51753e+05	2.14461e+06
332	1.73749e+06	3.47498e+06	2.88089e+06	1.94318e+06	34.0000	-8.01353e+05	2.12874e+06
333	1.76536e+06	3.53071e+06	2.85641e+06	2.07530e+06	36.0000	-7.51285e+05	2.11164e+06
334	1.79020e+06	3.58040e+06	2.82139e+06	2.20431e+06	38.0000	-7.01737e+05	2.09334e+06
335	1.81200e+06	3.62401e+06	2.77615e+06	2.32947e+06	40.0000	-6.52889e+05	2.07391e+06
336	1.83077e+06	3.66154e+06	2.72105e+06	2.45005e+06	42.0000	-6.04917e+05	2.05341e+06
337	1.84650e+06	3.69301e+06	2.65653e+06	2.56538e+06	44.0000	-5.57989e+05	2.03193e+06
338	1.85921e+06	3.71842e+06	2.58303e+06	2.67481e+06	46.0000	-5.12265e+05	2.00956e+06
339	1.86891e+06	3.73783e+06	2.50109e+06	2.77775e+06	48.0000	-4.67898e+05	1.98639e+06
340	1.87563e+06	3.75126e+06	2.41126e+06	2.87363e+06	50.0000	-4.25029e+05	1.96255e+06
341	1.87939e+06	3.75878e+06	2.31413e+06	2.96196e+06	52.0000	-3.83793e+05	1.93816e+06
342	1.88023e+06	3.76045e+06	2.21034e+06	3.04227e+06	54.0000	-3.44309e+05	1.91333e+06
343	1.87818e+06	3.75636e+06	2.10053e+06	3.11416e+06	56.0000	-3.06690e+05	1.88821e+06
344	1.87330e+06	3.74659e+06	1.98539e+06	3.17729e+06	58.0000	-2.71033e+05	1.86293e+06
345	1.86562e+06	3.73125e+06	1.86562e+06	3.23136e+06	60.0000	-2.37427e+05	1.83763e+06
346	1.85522e+06	3.71044e+06	1.74195e+06	3.27613e+06	62.0000	-2.05945e+05	1.81244e+06
347	1.84215e+06	3.68429e+06	1.61509e+06	3.31142e+06	64.0000	-1.76649e+05	1.78751e+06
348	1.82646e+06	3.65293e+06	1.48578e+06	3.33712e+06	66.0000	-1.49589e+05	1.76296e+06
349	1.80824e+06	3.61649e+06	1.35476e+06	3.35315e+06	68.0000	-1.24801e+05	1.73893e+06
350	1.78756e+06	3.57512e+06	1.22276e+06	3.35952e+06	70.0000	-1.02307e+05	1.71552e+06
351	1.76449e+06	3.52898e+06	1.09051e+06	3.35626e+06	72.0000	-8.21203e+04	1.69286e+06
352	1.73911e+06	3.47823e+06	9.58730e+05	3.34349e+06	74.0000	-6.42372e+04	1.67104e+06
353	1.71152e+06	3.42304e+06	8.28107e+05	3.32136e+06	76.0000	-4.86441e+04	1.65014e+06
354	1.68179e+06	3.36358e+06	6.99327e+05	3.29007e+06	78.0000	-3.53148e+04	1.63025e+06
355	1.65002e+06	3.30003e+06	5.73044e+05	3.24990e+06	80.0000	-2.42112e+04	1.61141e+06
356	1.61629e+06	3.23259e+06	4.49889e+05	3.20113e+06	82.0000	-1.52844e+04	1.59367e+06
357	1.58072e+06	3.16144e+06	3.30461e+05	3.14412e+06	84.0000	-8.47461e+03	1.57707e+06
358	1.54339e+06	3.08678e+06	2.15323e+05	3.07926e+06	86.0000	-3.71175e+03	1.56160e+06
359	1.50441e+06	3.00881e+06	1.05006e+05	3.00698e+06	88.0000	-9.16350e+02	1.54727e+06
360	1.46387e+06	2.92773e+06	1.29182e-09	2.92773e+06	90.0000	-3.07686e-10	1.53405e+06

Appendix E. Formula and variable ledger

Symbol	Meaning	Units or type
Q	Symmetric positive-definite phase-transition and kinetic-cost matrix	dimensionless 3 x 3 matrix
c_hat	Earth-CMB unit direction and axis B	unit vector
V_E	Earth-CMB velocity vector	m/s
e0	Transverse bar reference at phi = 0	unit vector
A0	Inner-axis reference at phi = 0	unit vector
phi	Outer phase omega_B t	radian
u(phi)	Positive endpoint radial direction	unit vector
u_phi	du/dphi	dimensionless derivative
B(phi)	u_phi^T Q V_E	m/s
F_net	Net force on two endpoint masses	N in native Q-metric normalization
F_parallel	Projection of F_net on c_hat	N
F_perp	Remainder perpendicular to c_hat	N
J_B	Complete outer-cycle impulse	N s
T_B	Outer period 2pi/omega_B	s
B_K	Cell3 ICRS basis matrix with columns a,b,c	native length

References and source ledger

#	Source	Use
1	K-Field Dual-Axis Kinetic Mass Analysis, archival computational edition, Unix epoch 1784741897.	Primary apparatus, Q matrix, one-degree force vectors, and analytic impulse theorem.
2	K-Field Dual-Axis Kinetic Mass CMB Traversal Geometry, archival computational edition, Unix epoch 1784745791.	Earth-center RA = 0 degree datum, absolute translated path, and Cell3 traversal coefficients.
3	Authoritative source machine-readable files listed in the accompanying JSON.	Numerical values and reproducible full-precision data.

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